

LOWER LIMB

Outline

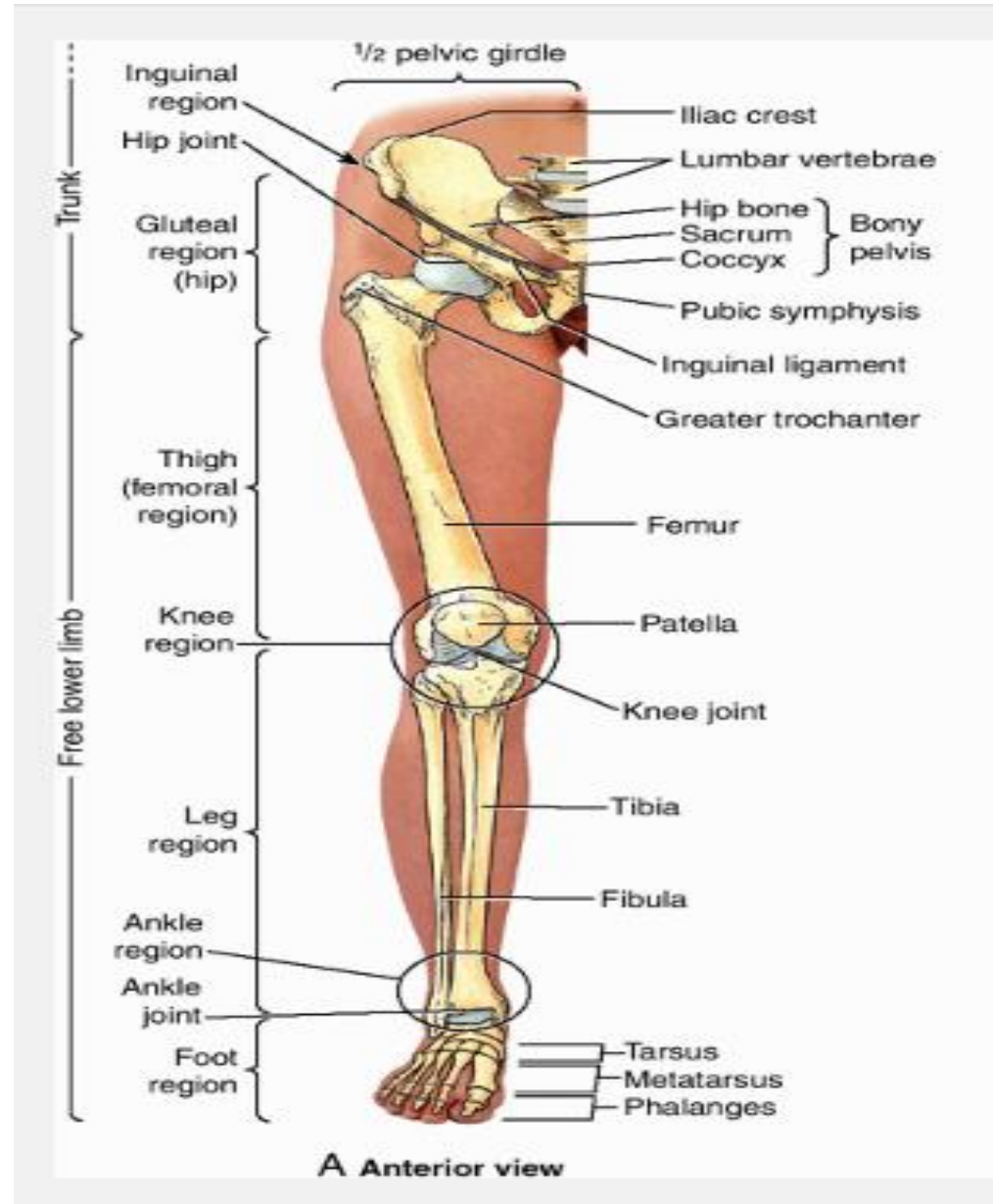
- Introduction
- Bones of the lower limb
- Superficial structures of lower limb
- Thigh and gluteal region
- Popliteal fossa
- Leg
- Foot
- Joints of lower limb

Introduction

- Function
 - Locomotion
 - Support body weight
 - Maintain balance
- Connected to trunk by pelvic girdle
 - Composed of sacrum, right and left hip bones joined anteriorly at pubic symphysis

Regions

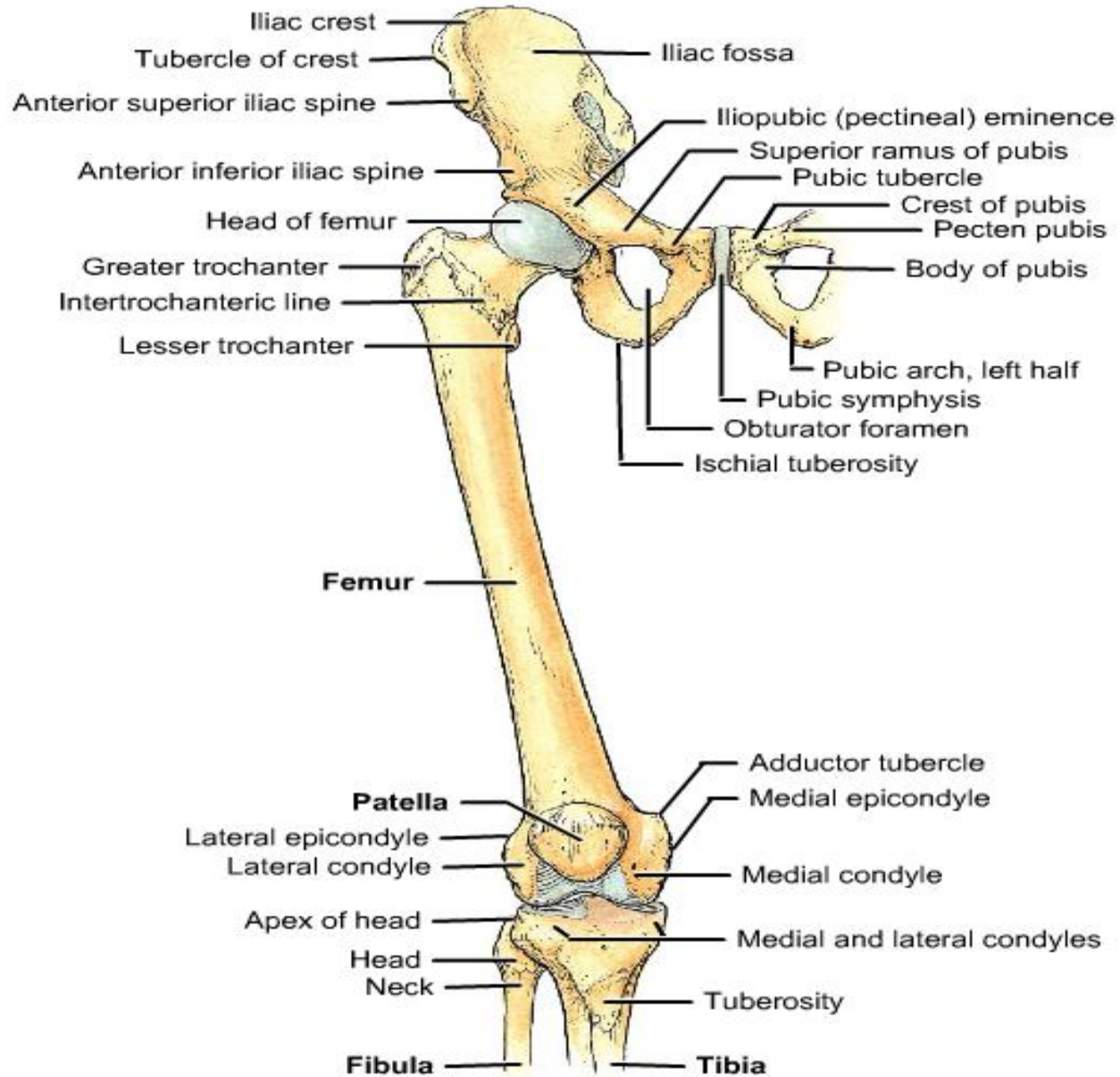
- Gluteal region: buttocks and hip region
- Thigh or femoral region
- Knee region
- Leg
- Ankle
- Foot

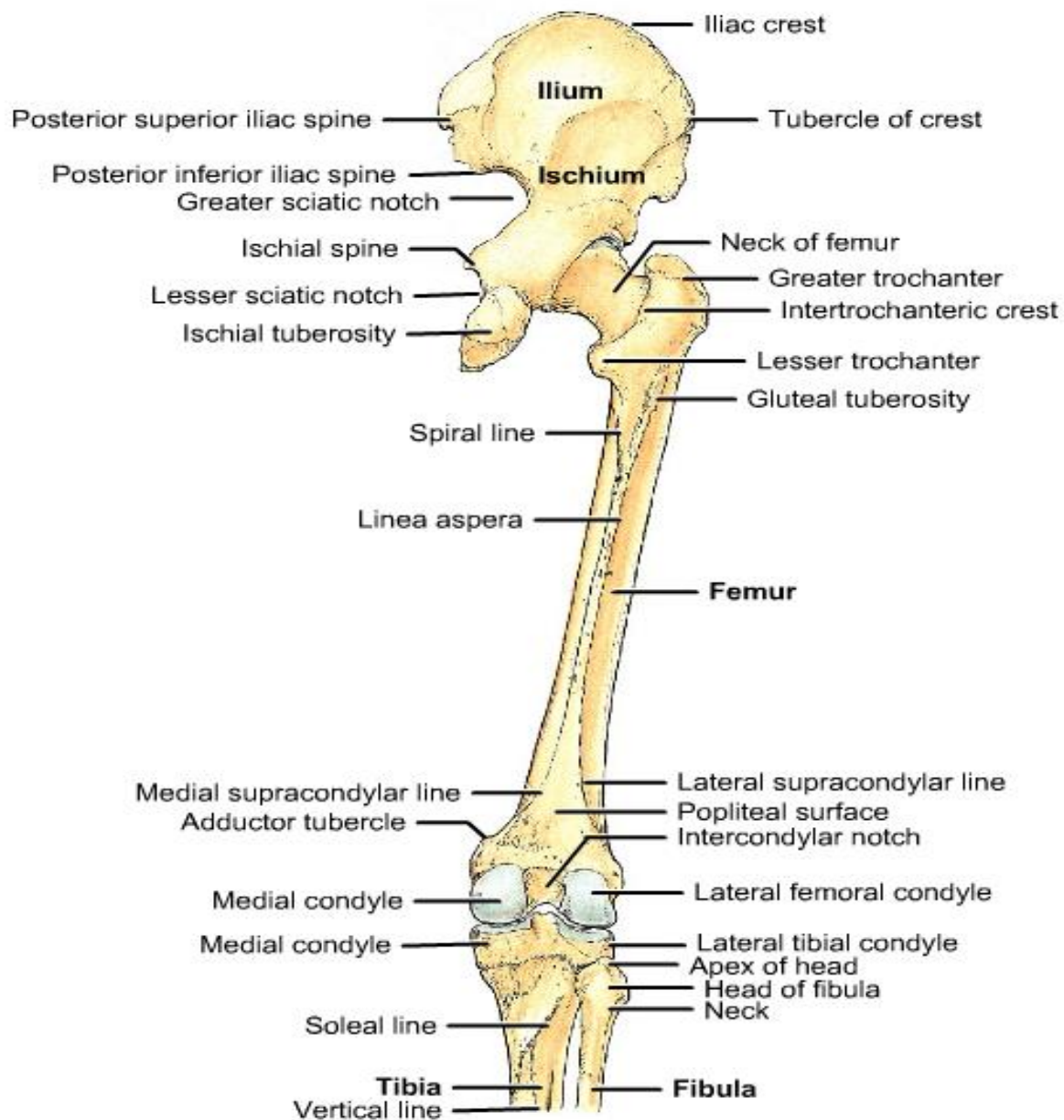


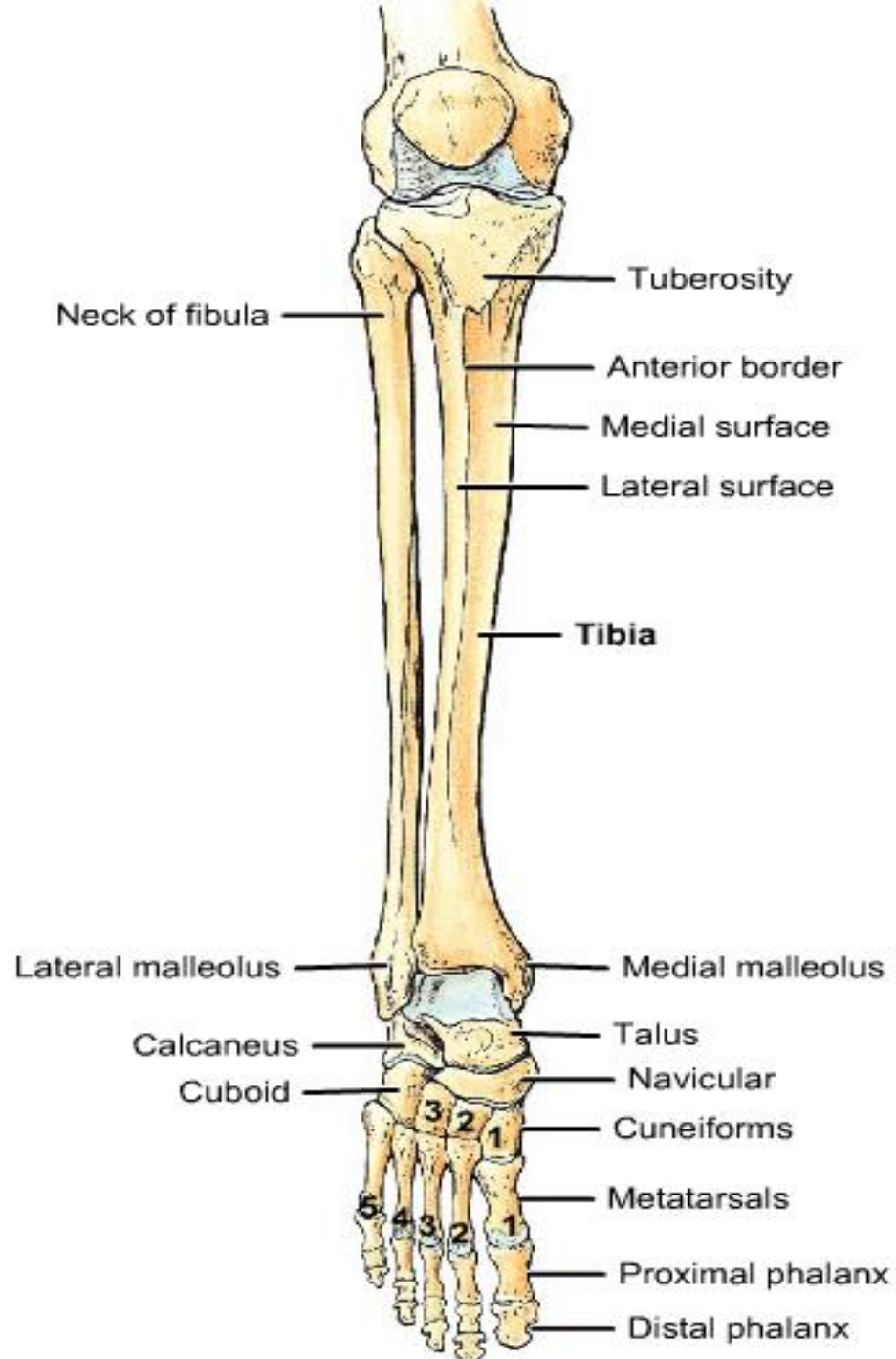
Bones of lower limb

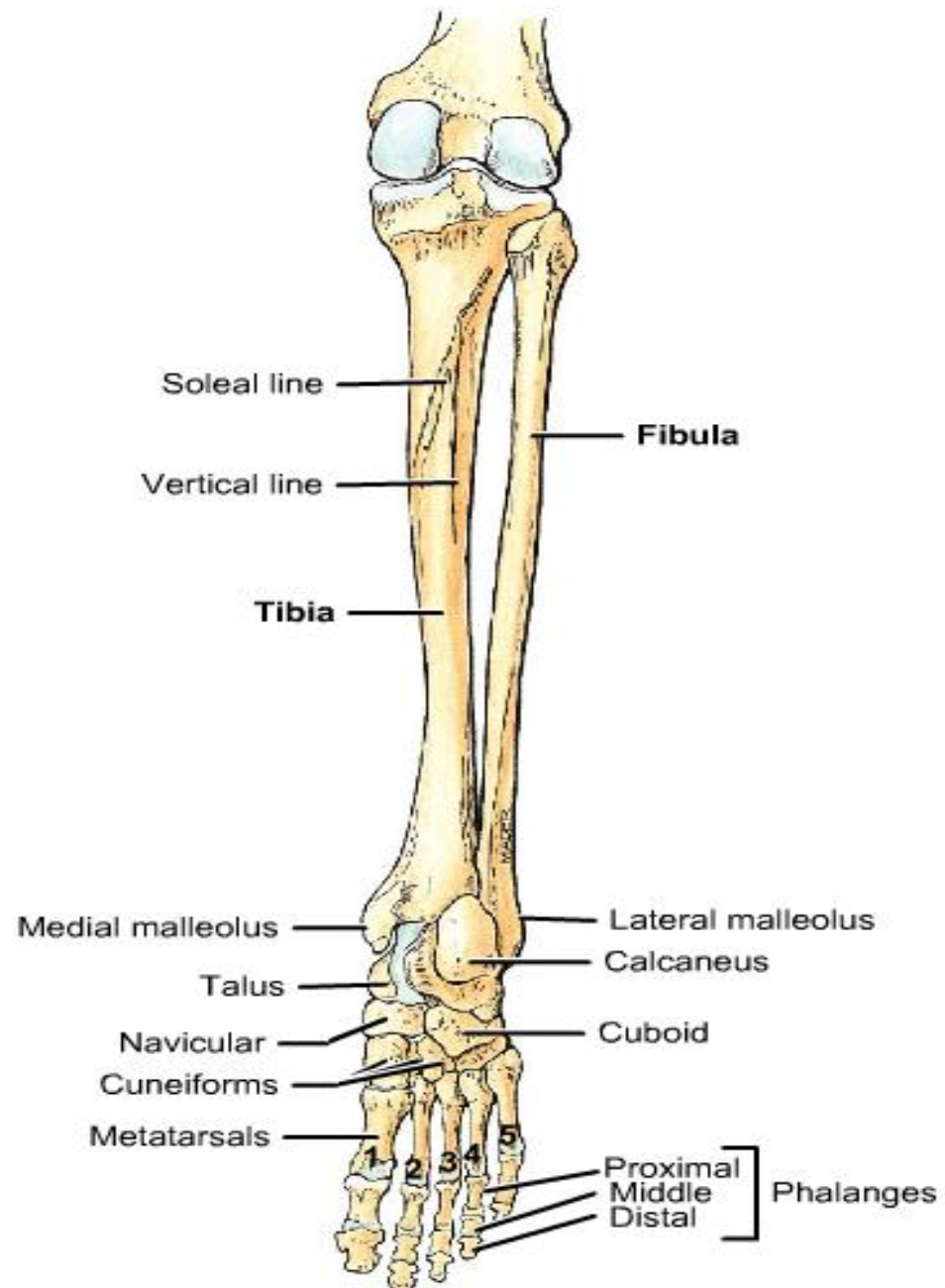
Bone.....cont'd

- Hip region
 - Hip bone
- Thigh
 - Femur
- Knee
 - Patella
- Leg
 - Tibia
 - Fibula
- Foot
 - 7 Tarsal bones
 - 5 Metatarsal bones
 - 14 Phalanges









Pelvic Girdle

Basin-shaped ring of bones that connects the vertebral column to the femurs in the thighs.

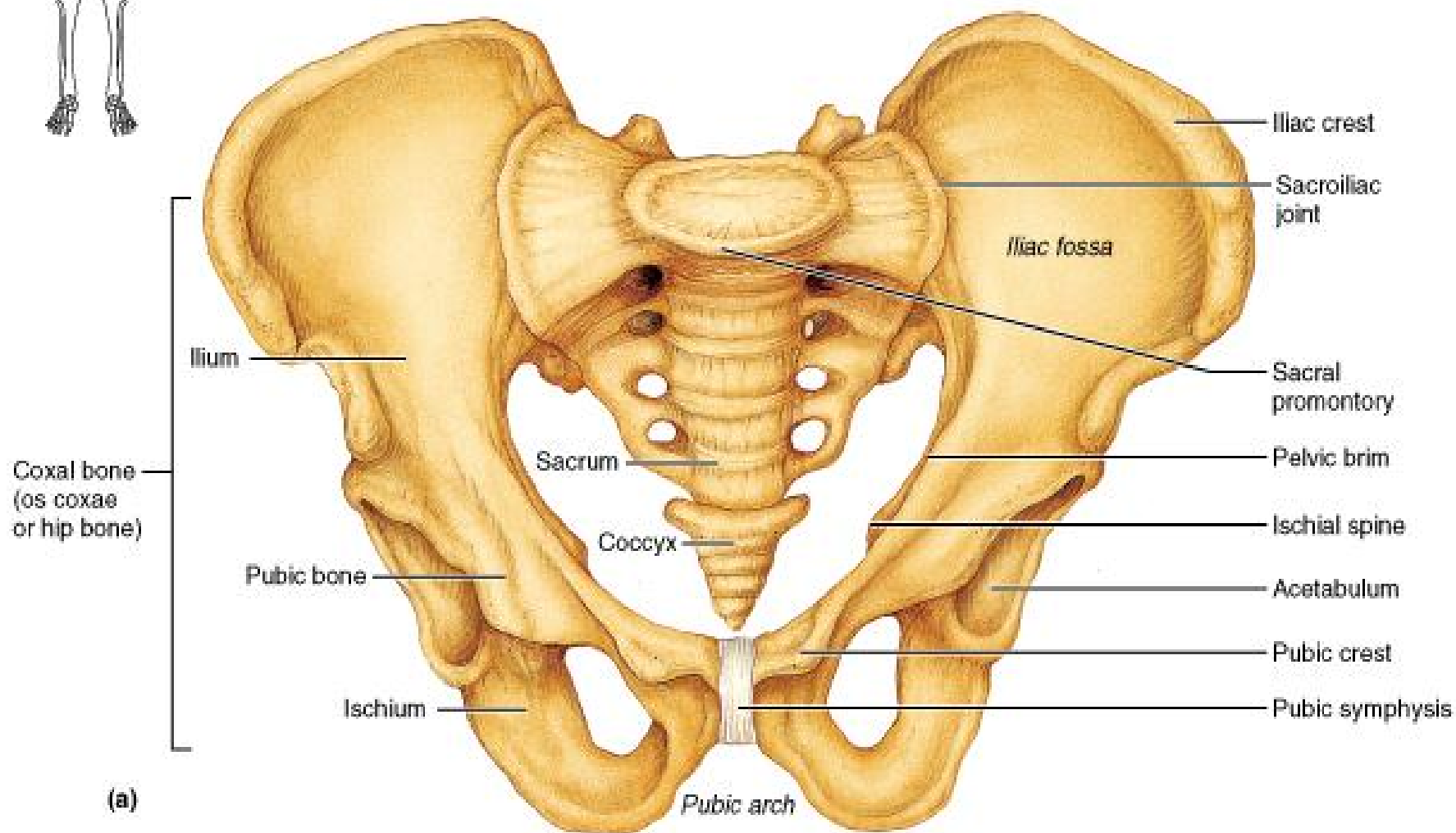
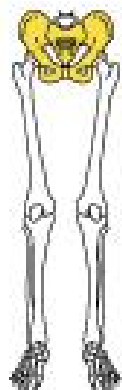
Functions

- Bear the weight of the upper body (sitting and standing)
- Transfer the weight of the upper body from the axial to the lower appendicular skeleton.
- Provide attachment for the powerful muscles of locomotion and posture.
- Withstand compression and other forces resulting from its support of body weight
- Contain and protect the pelvic and abdominal viscera.

The bony pelvis is formed by 4 bones united by 4 joints

- Bones: 2 hip bones, sacrum and coccyx

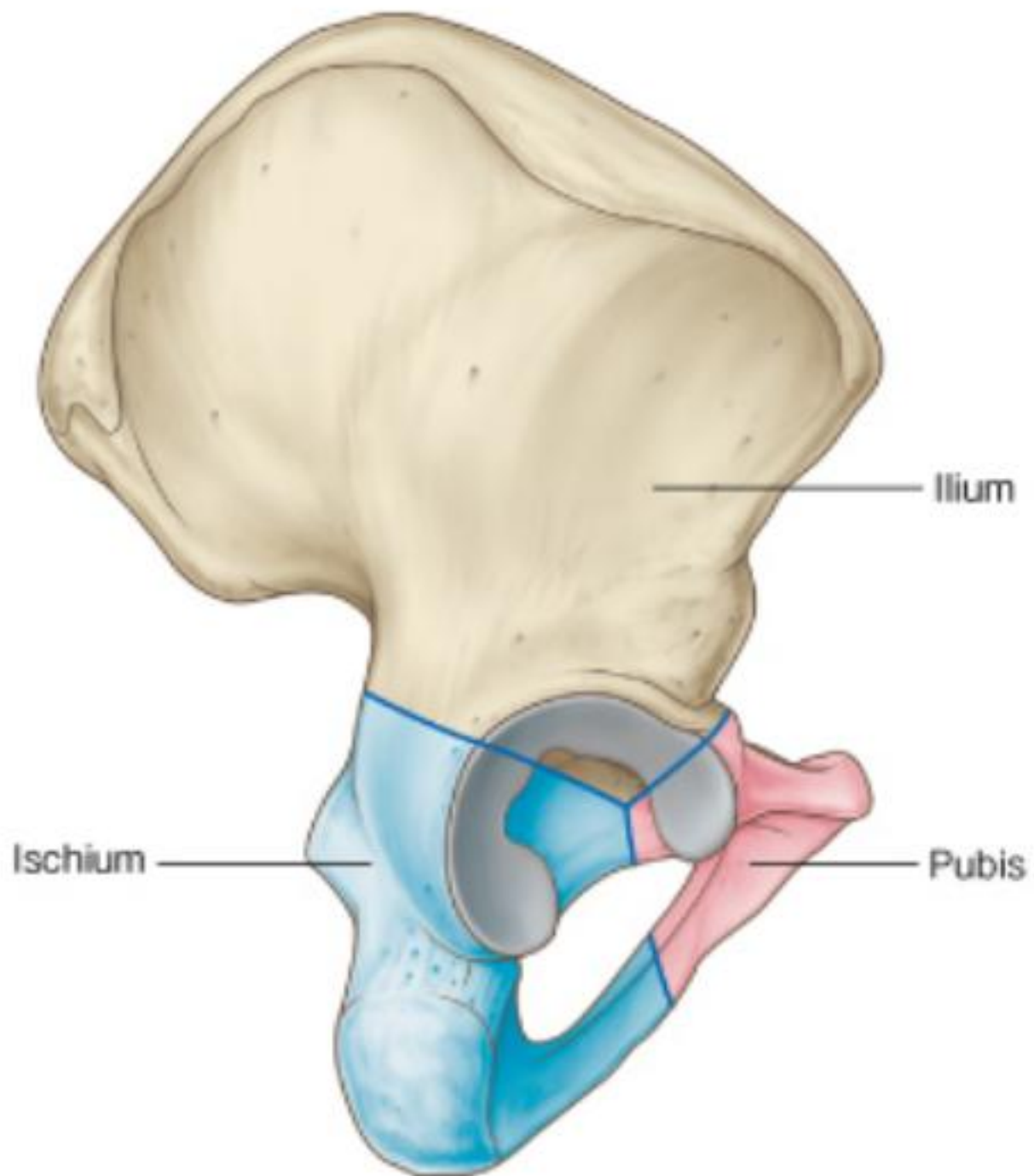
- Joints: 2 sacroiliac joints, pubic symphysis and sacro-coccygeal joint



Hip bones

- Attaches the lower limbs to axial skeleton
- Transfers the weight of the body to lower limbs
- Supports the visceral organs of the pelvis
- Secured by strong ligaments
- Less range of motion in all planes of movement
- The two hip bones are joined at the pubic symphysis anteriorly and to the sacrum posteriorly at the sacroiliac joints to form a bony ring, the pelvic girdle

- Each hip bone (os coxae) is formed by 3 bones fusing at the acetabulum (a cup-like articular depression on lateral aspect for the head of the femur) by a y-shaped cartilage
- Begin to fuse at 15-17 years and complete at 20-25 years of age
- The 3 bones are:
 - Ilium
 - Ischium
 - Pubis



Ilium

- The superior, flattened, fan-shaped part of the hip bone
 - the ala of the ilium represents the spread of the fan; and the body of the ilium, the handle of the fan
- Located superior to the acetabulum
- Body
 - forms the superior part of the acetabulum
 - joins ischium and pubis at acetabulum
- Ala (wing)
 - bordered superiorly by iliac crest
 - dorsum feature anterior, posterior and inferior gluteal lines (origins of gluteus minimus, medius and maximus muscles)

➤ Iliac crest

- the rim of the ilium
- has a curve that follows the contour of the ala between the anterior and the posterior superior iliac spines
- a line joining the most superior point of the bilateral iliac crests will cross the 4th lumbar spine (L4)
- the anterior concave part of the ala forms the iliac fossa

➤ Landmarks:

- Anterior superior iliac spine
- Anterior inferior iliac spine
- Posterior superior iliac spine
- Posterior inferior iliac spine, Greater sciatic notch

Ischium

- Posteroinferior part of hip bone
- has a body and a ramus

○ Body

- forms the posterior part of the acetabulum
- joins ilium and superior ramus of pubis to form acetabulum

○ Ramus

- fuses with the inferior ramus of pubis
- forms part of the inferior boundary of the obturator foramen

Landmarks

- Ischial tuberosity
 - large posteroinferior protuberance of the ischium
 - supports body during sitting
- Ischial spine
 - small pointed posterior projection near the junction of the ramus and body
- Lesser sciatic notch

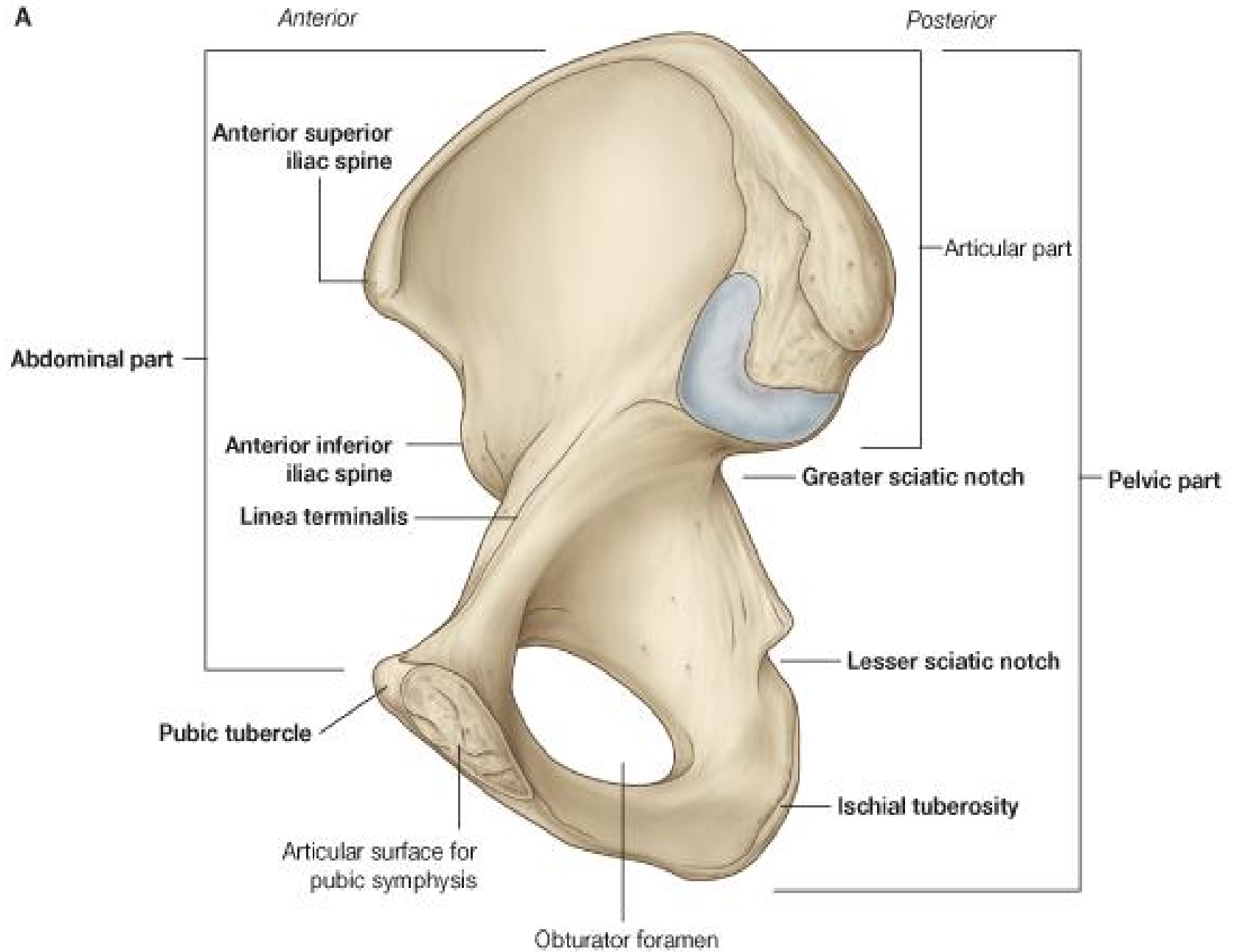
Pubis

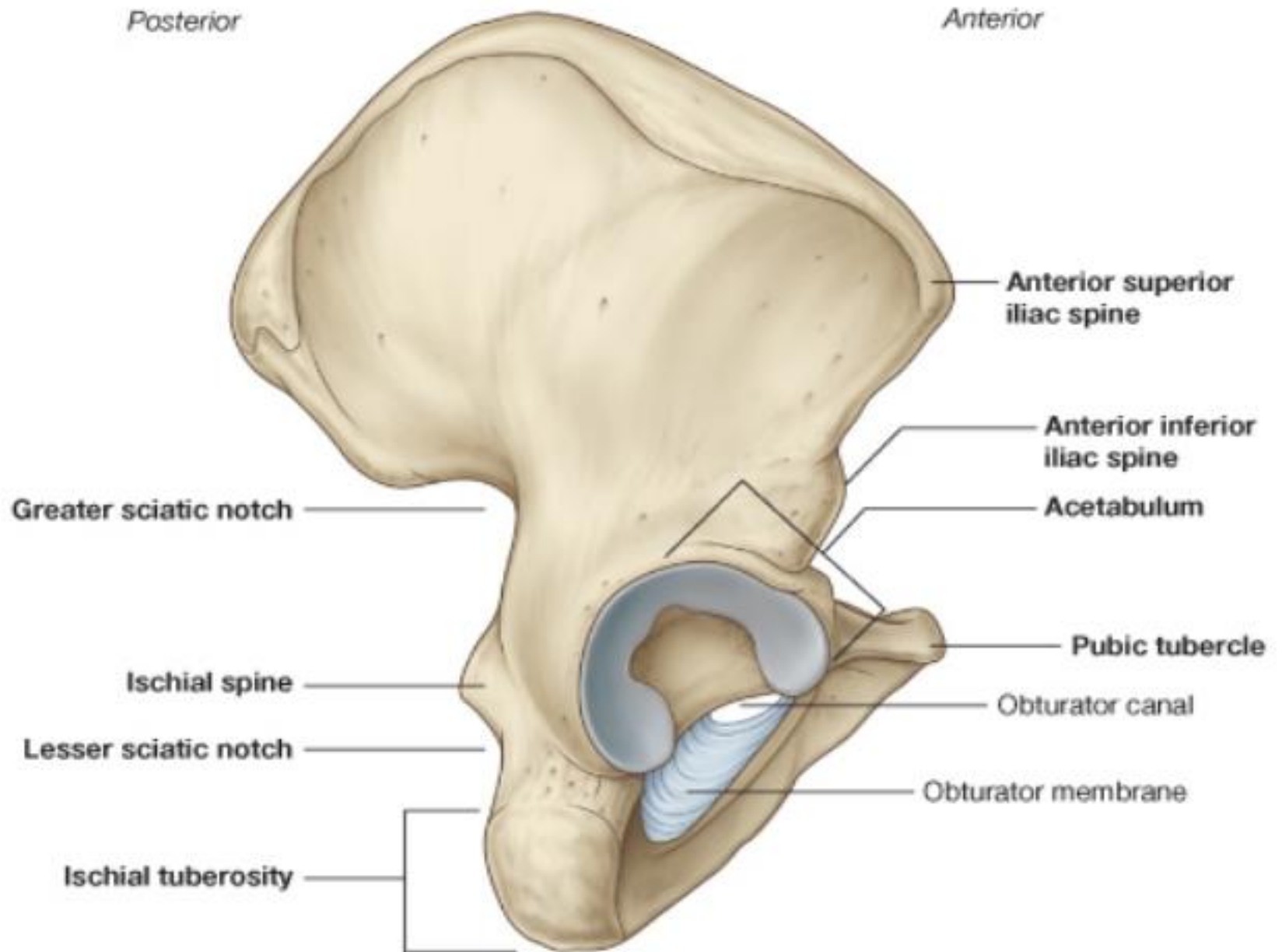
- Anteromedial part of hip bone
- Forms anterior part of the acetabulum
- Angulated bone; has two rami (inferior & superior) and body
- **Body:** has a symphysis surface for articulation with the contralateral pubis
- **Rami**
 - superior pubic ramus: forms anterior part of acetabulum
 - inferior pubic ramus: forms part of the inferior boundary of the obturator foramen

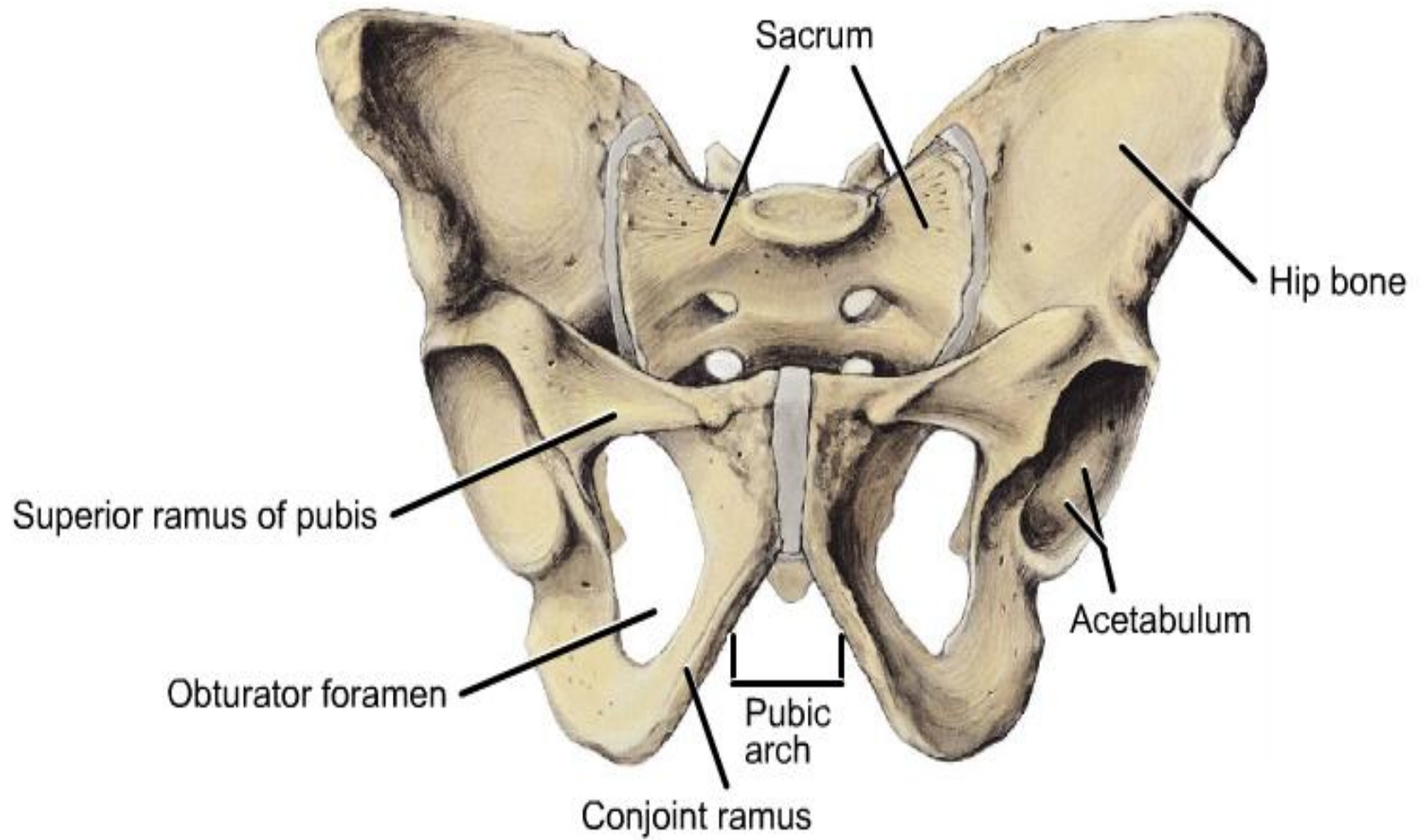
- Landmarks
 - Pecten pubis or pectineal line
 - oblique ridge on superior pubic ramus on its superior aspect
 - pubic crest
 - thickening on the anterior part of the body of the pubis
 - ends laterally as a swelling - pubic tubercle

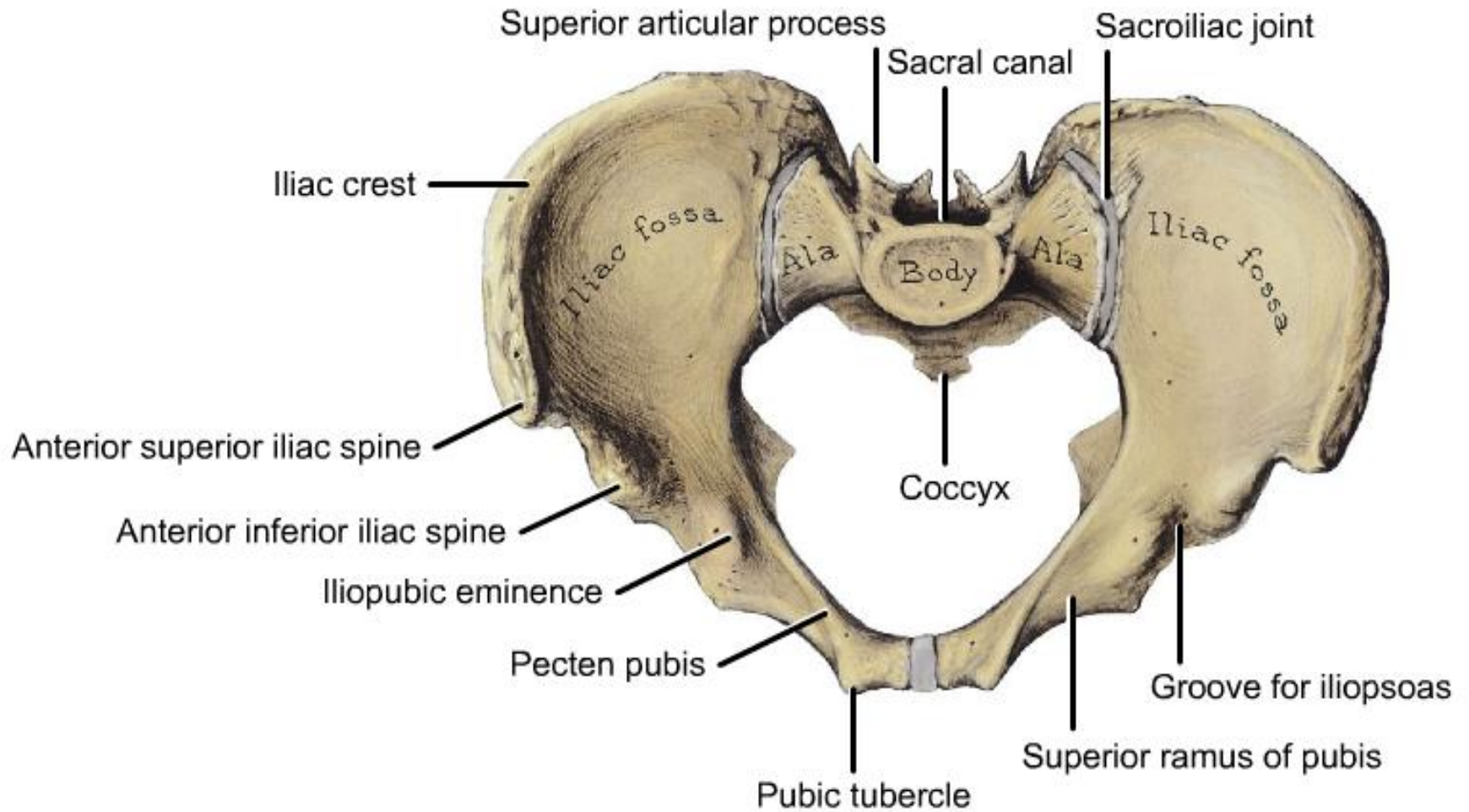
- Pubic arch (sub pubic angle)
 - formed by the ischio-pubic rami (conjoined inferior rami of the pubis and ischium) of the two sides
 - these rami meet at the pubic symphysis
 - their inferior borders define the subpubic angle
 - the distance between right and left ischial tuberosities
 - measured with the fingers in the vagina during a pelvic examination

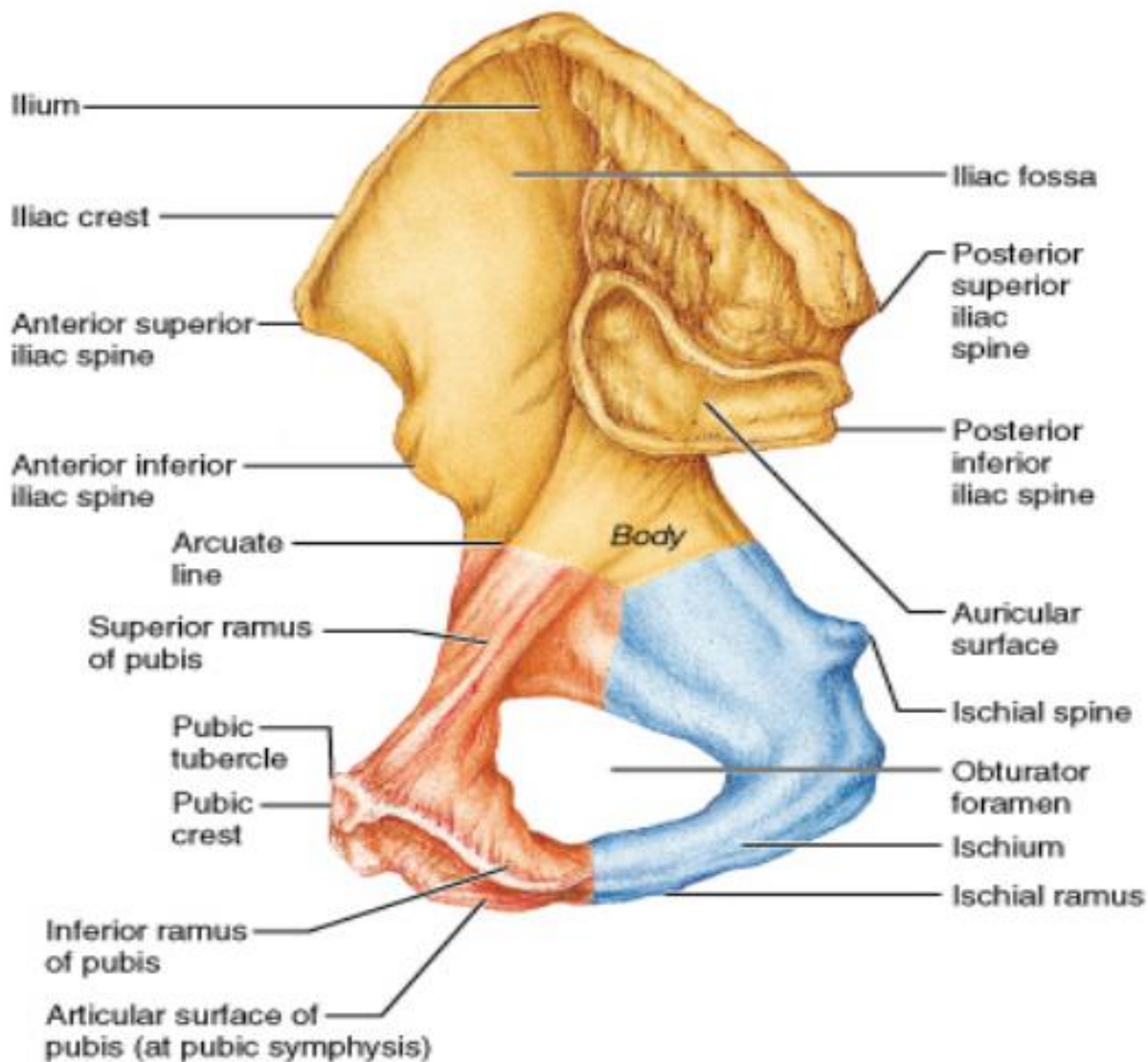
A



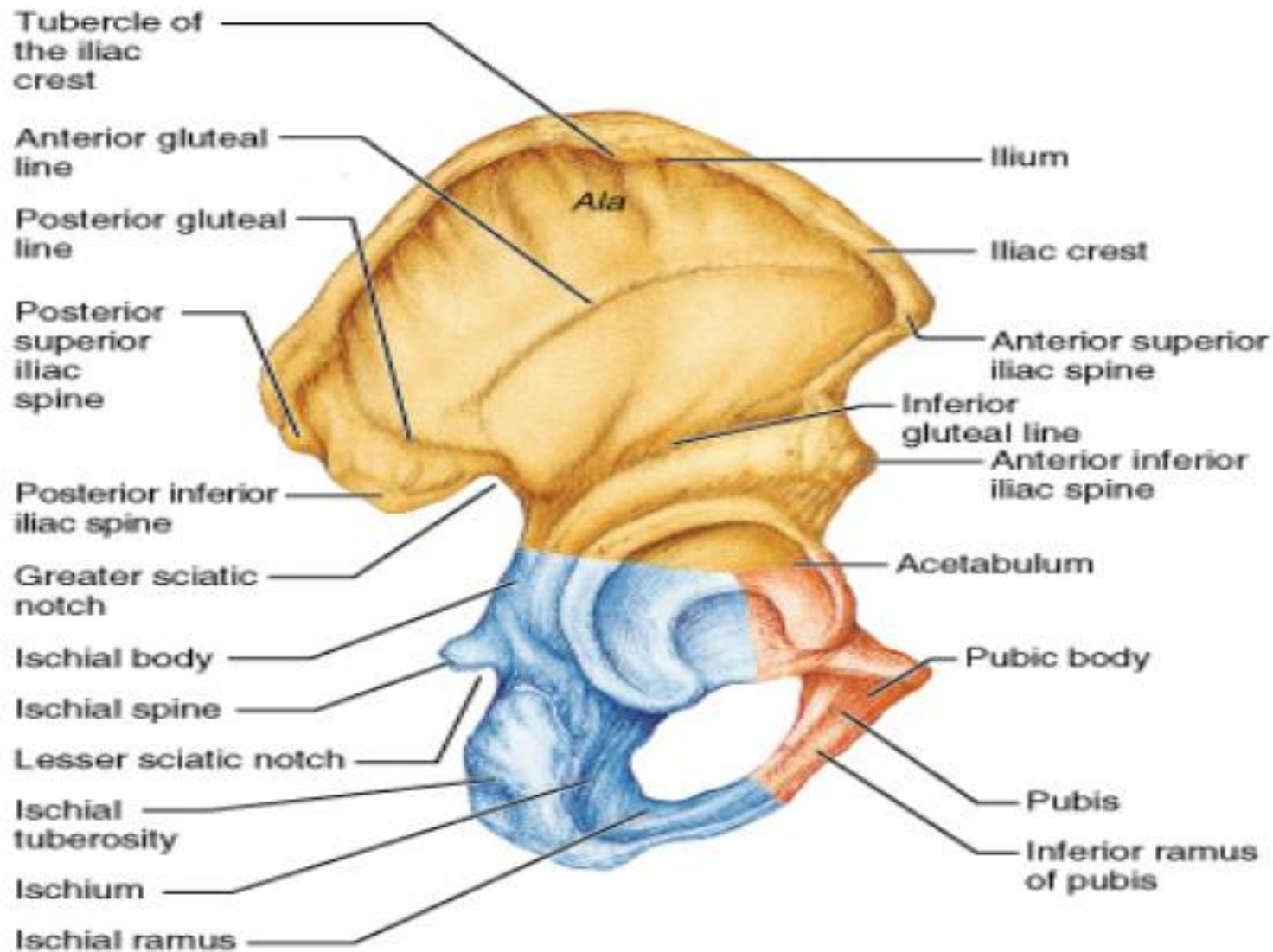








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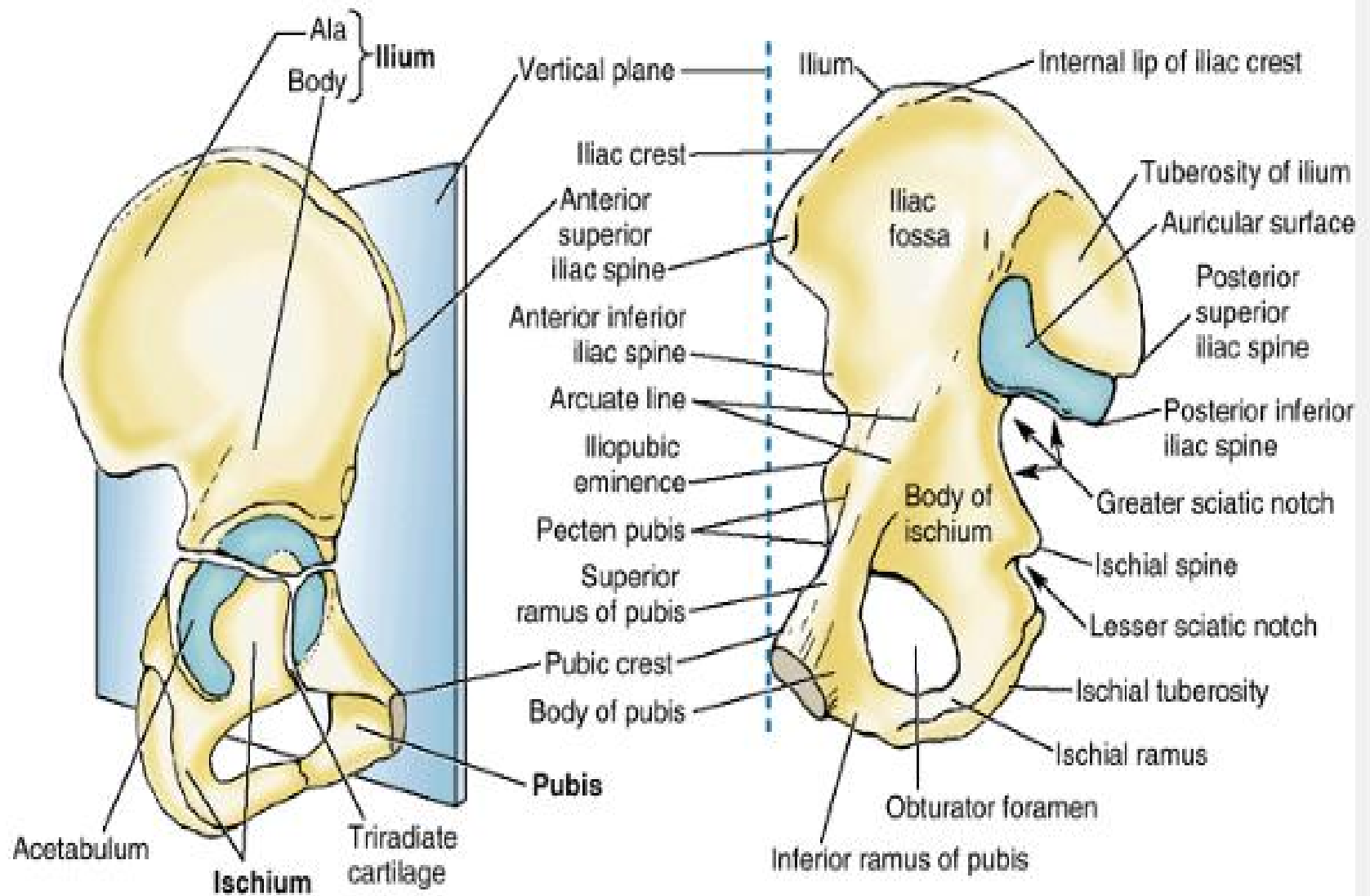


(b)

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Hip bone in anatomical position

- To place the hip bone in the anatomical position, manipulate
 - the acetabulum faces laterally and slightly anteriorly
- When the hip bone is in the anatomical position:
 - Anterior superior iliac spine (ASIS) and anterosuperior aspect of the pubis lie in the same vertical plane
 - Symphysial surface of the pubis is vertical, parallel to the median plane
 - Internal aspect of the body of the pubis faces almost directly superiorly
 - Acetabulum faces inferolaterally, with the acetabular notch directed inferiorly
 - Obturator foramen lies inferomedial to the acetabulum



B Lateral view

C Medial view

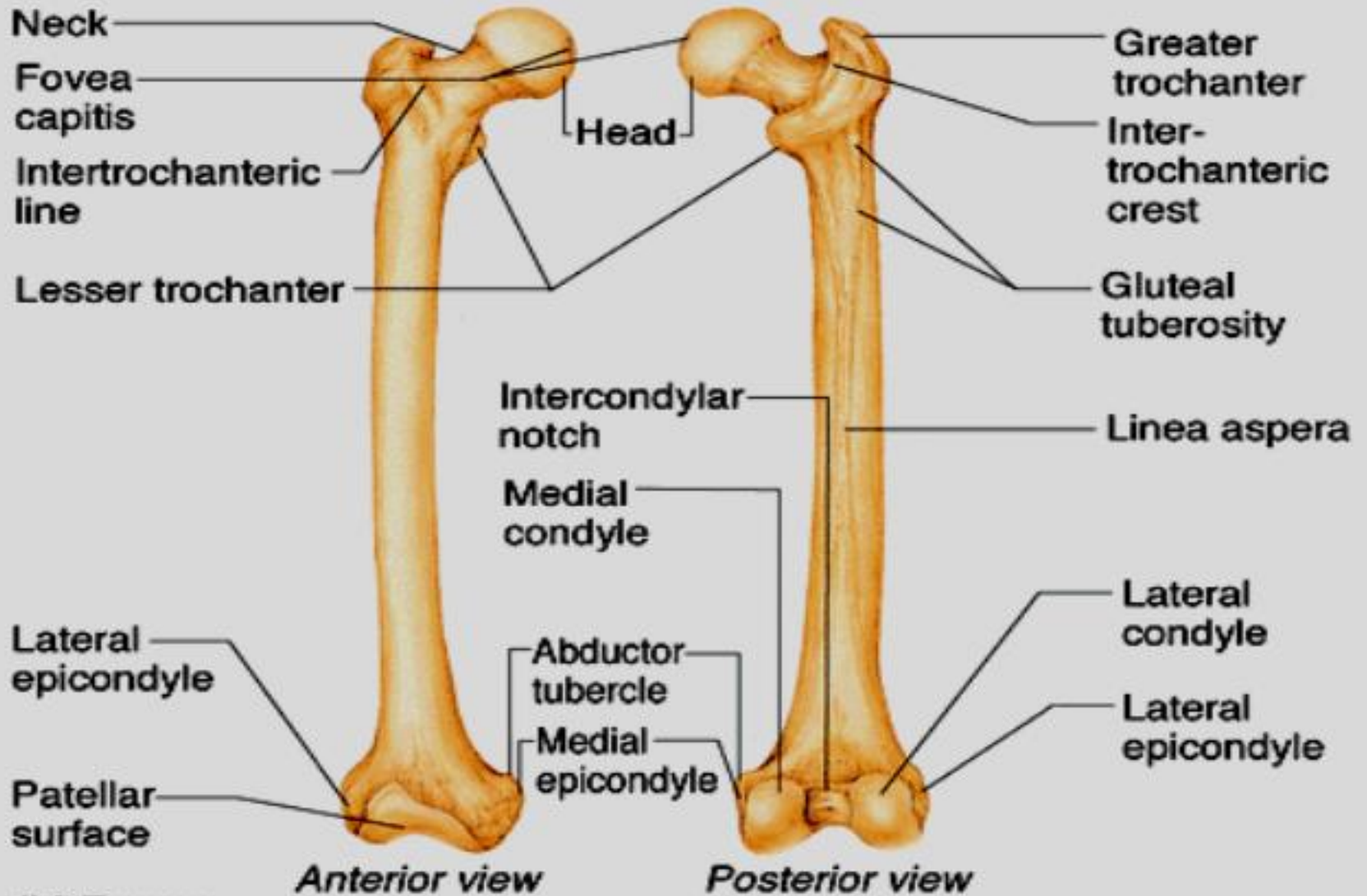
Fractures of hip bone

- Commonly referred to as pelvic fractures
- The term hip fracture is most commonly applied to fractures of the femoral heads, neck, or trochanters
- Avulsion fractures of the hip bone
 - may occur during sports that require sudden acceleration or deceleration forces
 - A small part of the bone with a piece of tendon or ligament attached is avulsed (torn away) for example, the anterior superior and inferior iliac spines, the ischiopubic rami, and the ischial tuberosities

Femur

- The longest, heaviest & strongest bone in the body
- To support the erect bipedal posture, femurs are obliquely placed
- Proximal end
 - Head
 - rounded
 - covered with articular cartilage except a medial pit, fovea for ligament of head
 - articulates with acetabulum
 - Neck
 - trapezoidal, narrow end supports head and broad base continuous with shaft

- Trochanters
 - where neck joins shaft are 2 blunt elevations - trochanters (greater & lesser) and intertrochanteric line (anteriorly) intertrochanteric crest (posteriorly) in between
- Shaft
 - possesses linea aspera; double edge ridge on posterior aspect
- Distal end
 - has medial & lateral condyles
 - articulates with tibial condyle to form knee joint
 - patellar surface for articulation with the patella
- Femoral fracture
 - neck is the most commonly fractured site



(b) Femur

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Patella (knee cap)

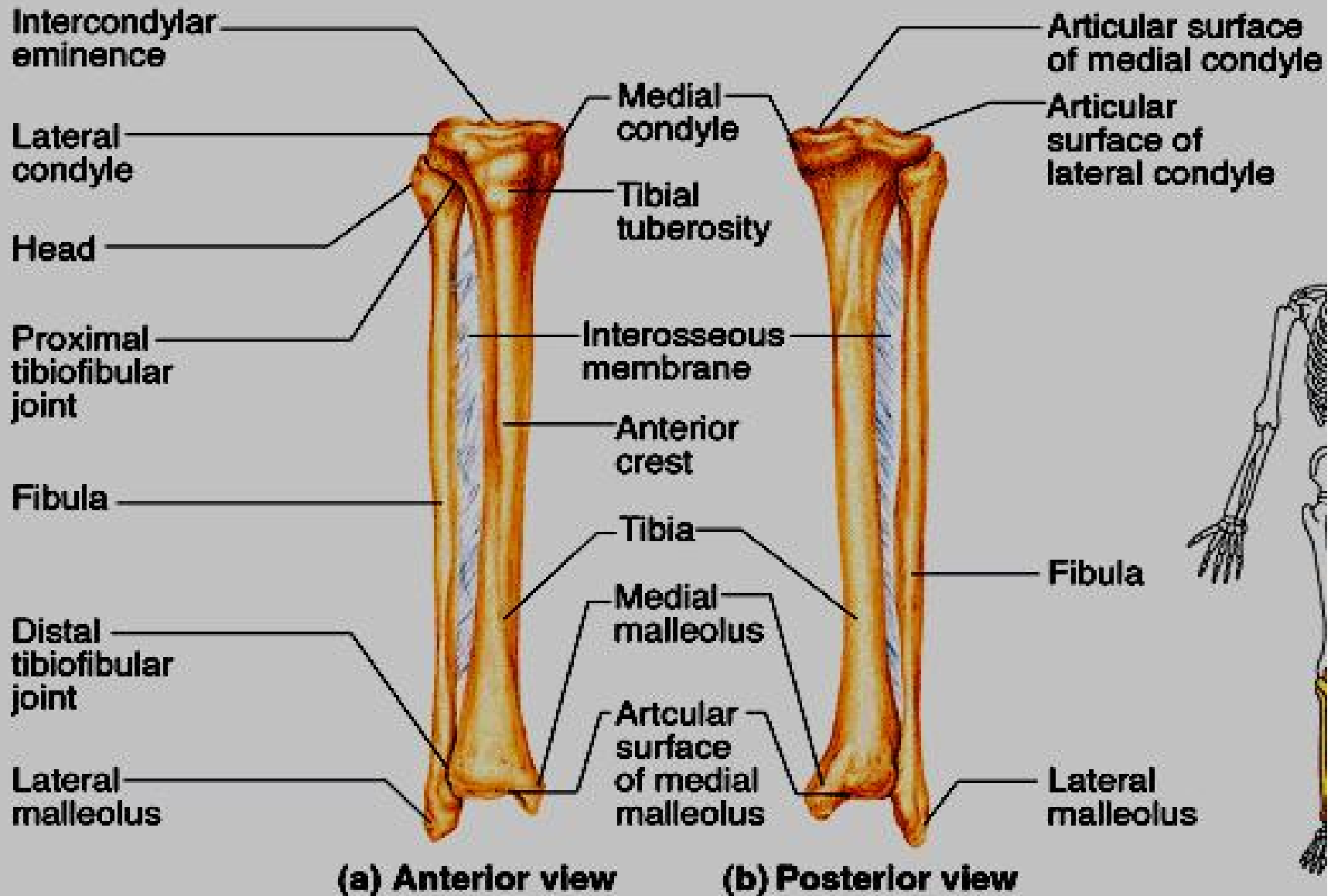
- Large sesamoid bone, formed intratendinously
- Located anterior to knee joint
- Triangular in shape
- Articulate with patellar surface of femur
- Anterior surface is convex
- Base – superior border
- Lateral and medial border converge to form apex

Tibia

- Large, weight bearing
- Articulate
 - proximally with condyles of femur at knee joint
 - distally with the talus of ankle
 - laterally at its proximal and distal ends with fibula
- has medial & lateral condyles and tibial tuberosity
- Medial malleolus - inferiorly directed projection from medial side of distal end
- Shaft is narrowest at the junction of its inferior and middle thirds, which is the most frequent site of fracture.

Fibula

- Long, slender bone on the lateral side of leg
- Serves mainly for muscle attachment than for support
- has head and neck
- Distal end enlarges to form lateral malleolus
- Fractures commonly occur just proximal to lateral malleolus and often are associated with dislocation of ankle joint
- Fibula is a common source of bone for grafting



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Bones of Foot

Tarsal Bones

- 7 bones which form the ankle joint; talus, calcaneus, navicular, 3 cuneiforms & cuboid
- Only talus articulate with leg bones
- Calcaneus
 - Heel bone
 - Largest and strongest bone
 - Articulate with talus and cuboid

- Talus
 - Rests on anterior 2/3 of calcaneus
 - Superior surface, trochlea of talus, bears weight of the body and articulates with two malleoli
 - Head articulate with navicular
- Navicular
 - Flattened, boat shaped
 - Located between talar head and cuneiforms
- Cuboid
 - Most lateral
- Cuneiforms: medial, intermediate and lateral
 - Wedge shaped
 - Each articulate with navicular posteriorly and metatarsal anteriorly

- Metatarsal Bones

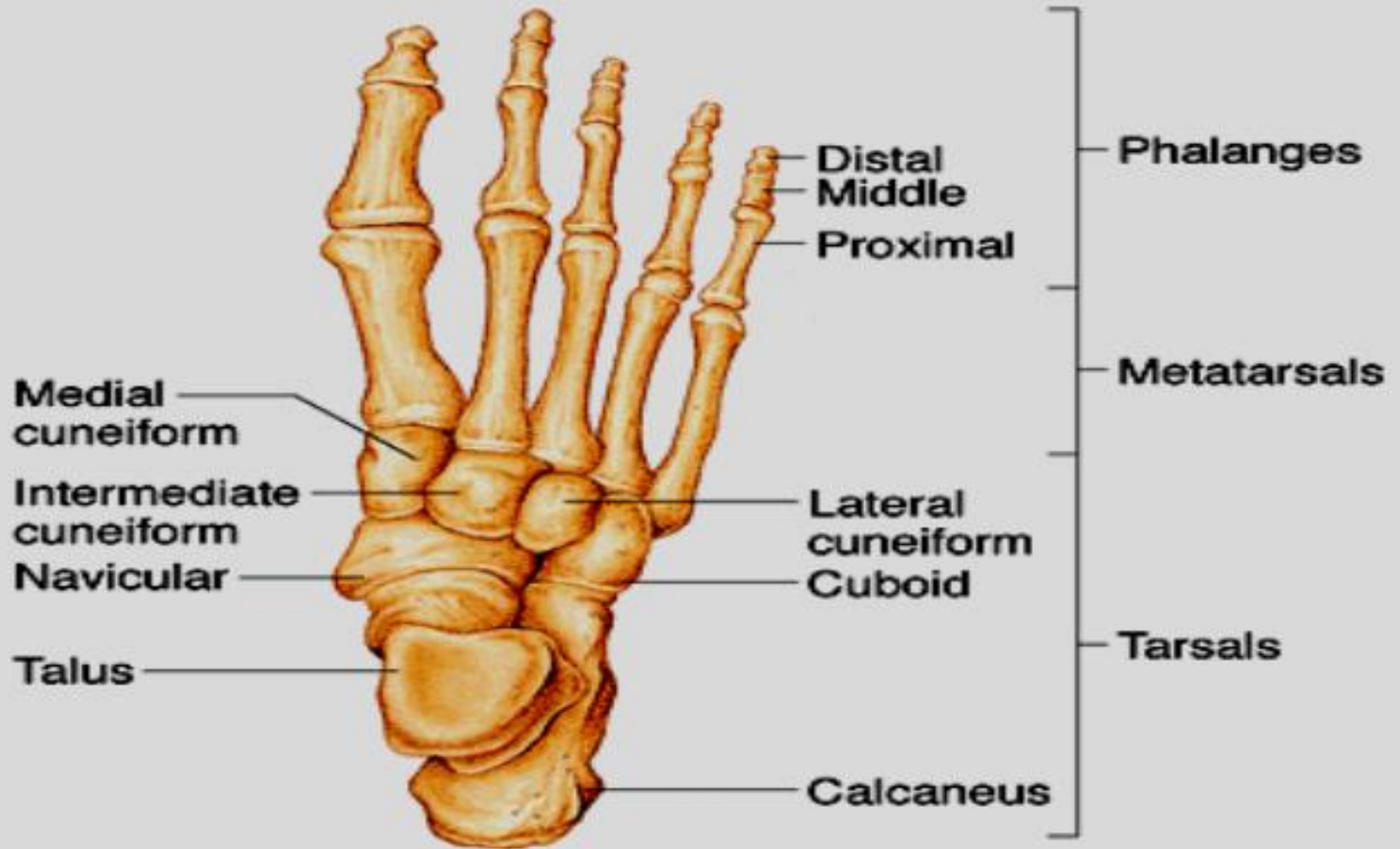
- 5 bones which form middle part of the foot
- Each bone has base, body & head
- Bases articulate with cuneiform and cuboid bones
- Heads articulate with proximal phalanges

- Phalanges

- 14 bones
- Lateral four digits have proximal, middle & distal phalanx
- Great toe (hallux) has only proximal & distal phalanx
- Each phalanx has base, body & head

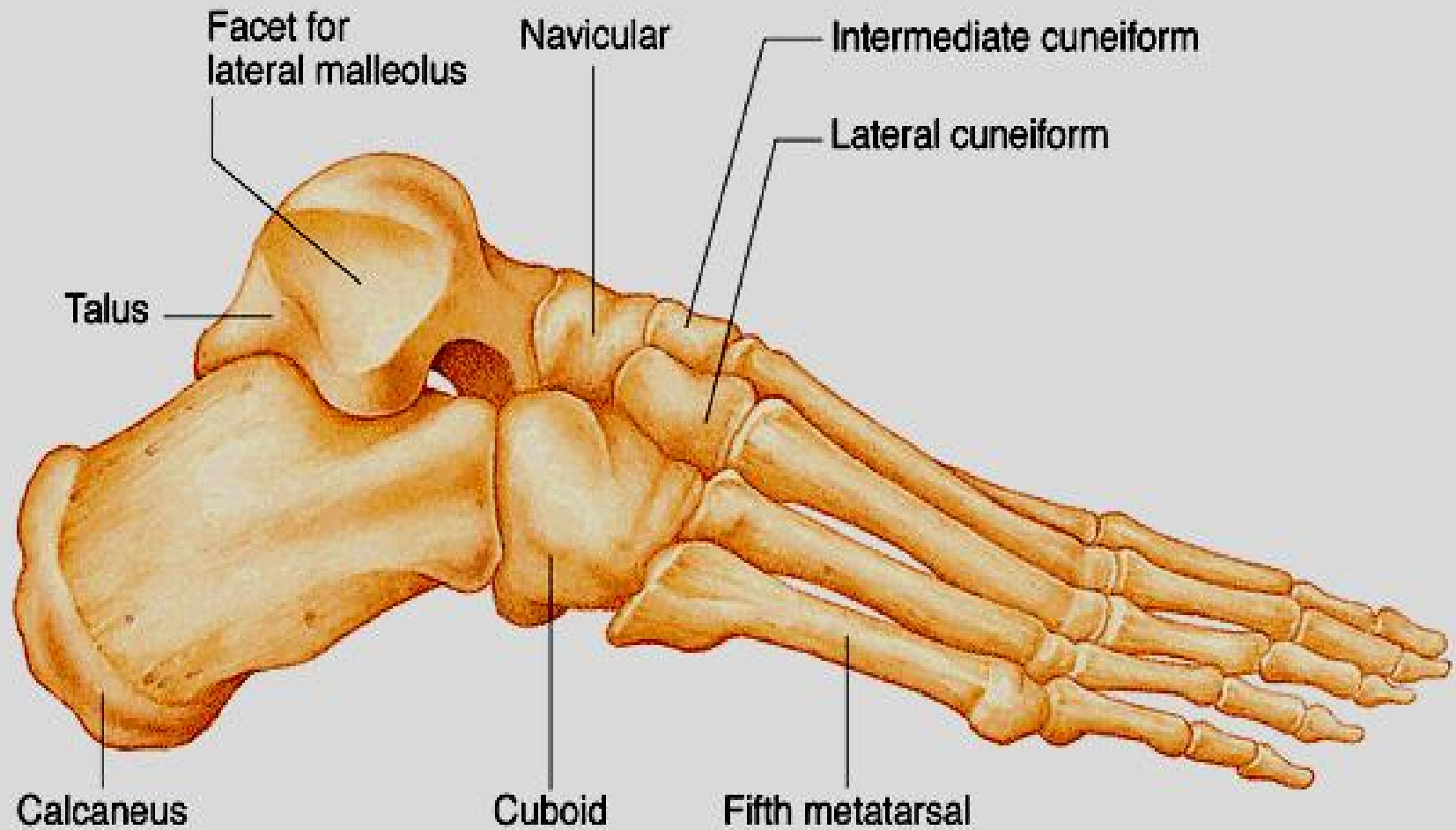
Fractures of foot bones

- Calcaneal fractures
 - occur in people who fall on their heels
 - Usually the bone breaks into several fragments (comminuted fracture) that disrupt the subtalar joint, where the talus articulates with the calcaneus
- Fractures of the talar neck
 - may occur during severe dorsiflexion of the ankle
- Metatarsal and phalangeal fractures
 - usually occur when a heavy object falls on the foot or when the foot is run over by a wheel bearing a heavy weight



(a) Superior view

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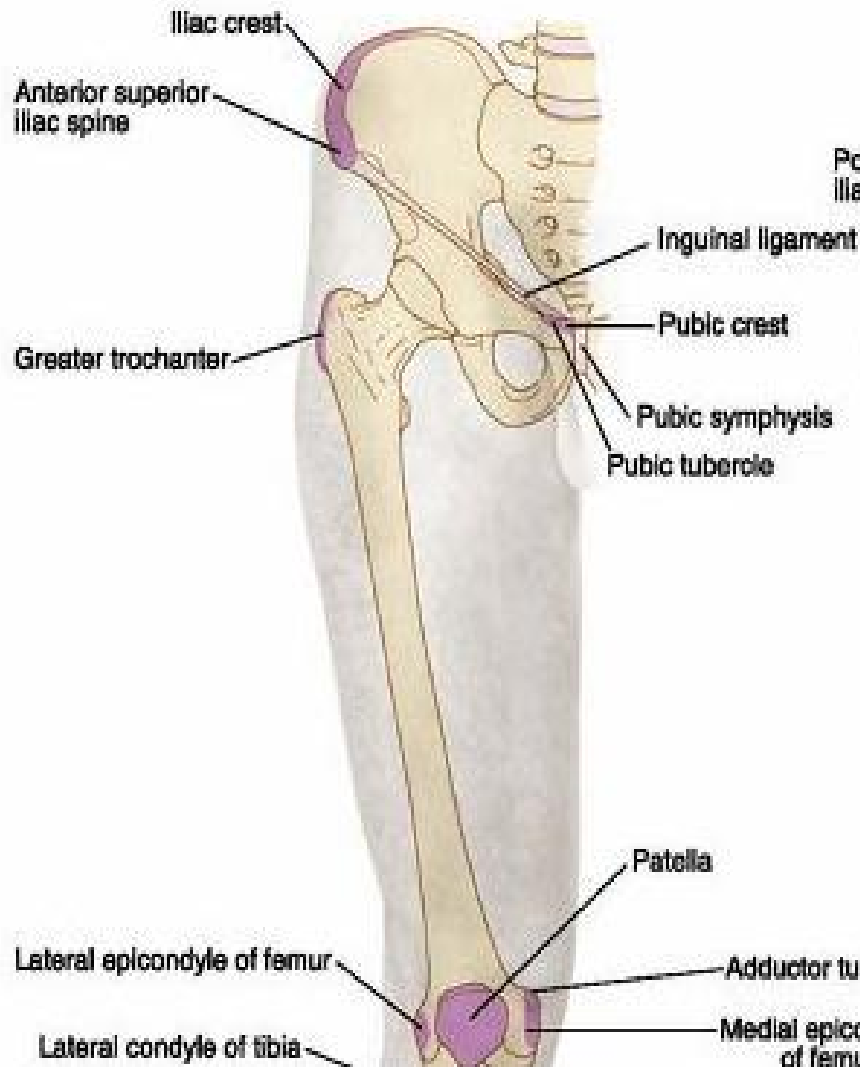
(c) Lateral view

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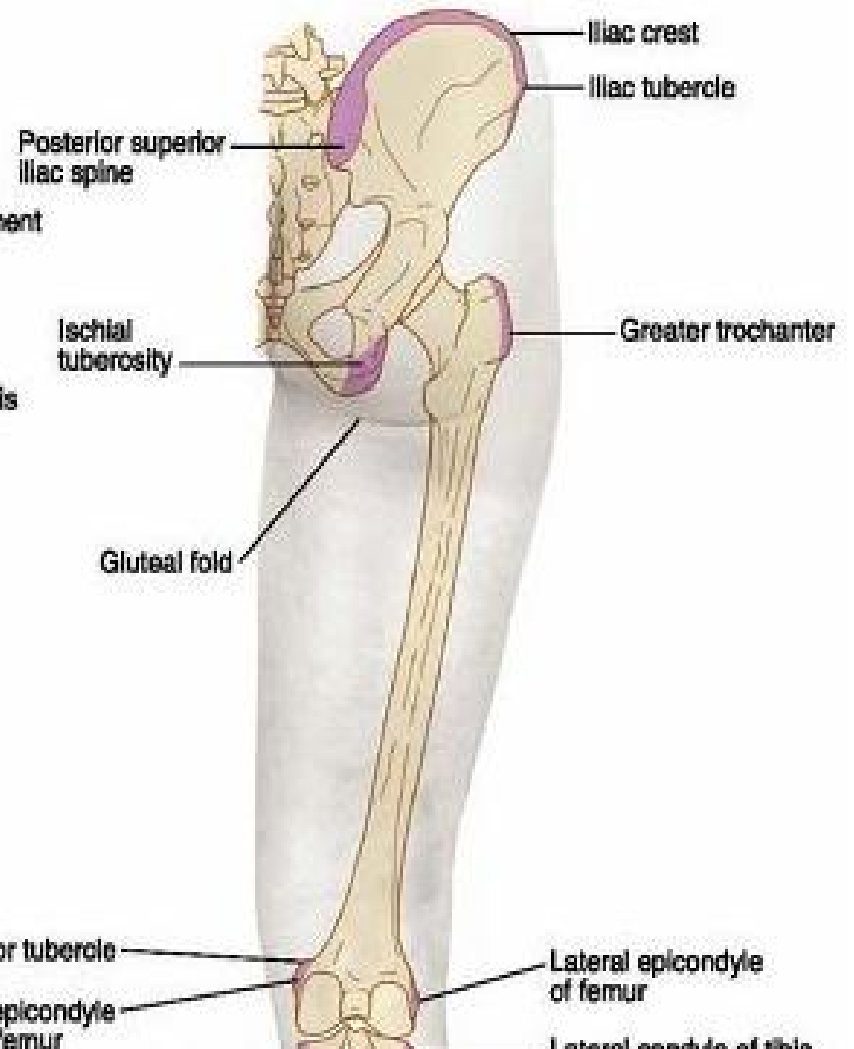
Surface anatomy

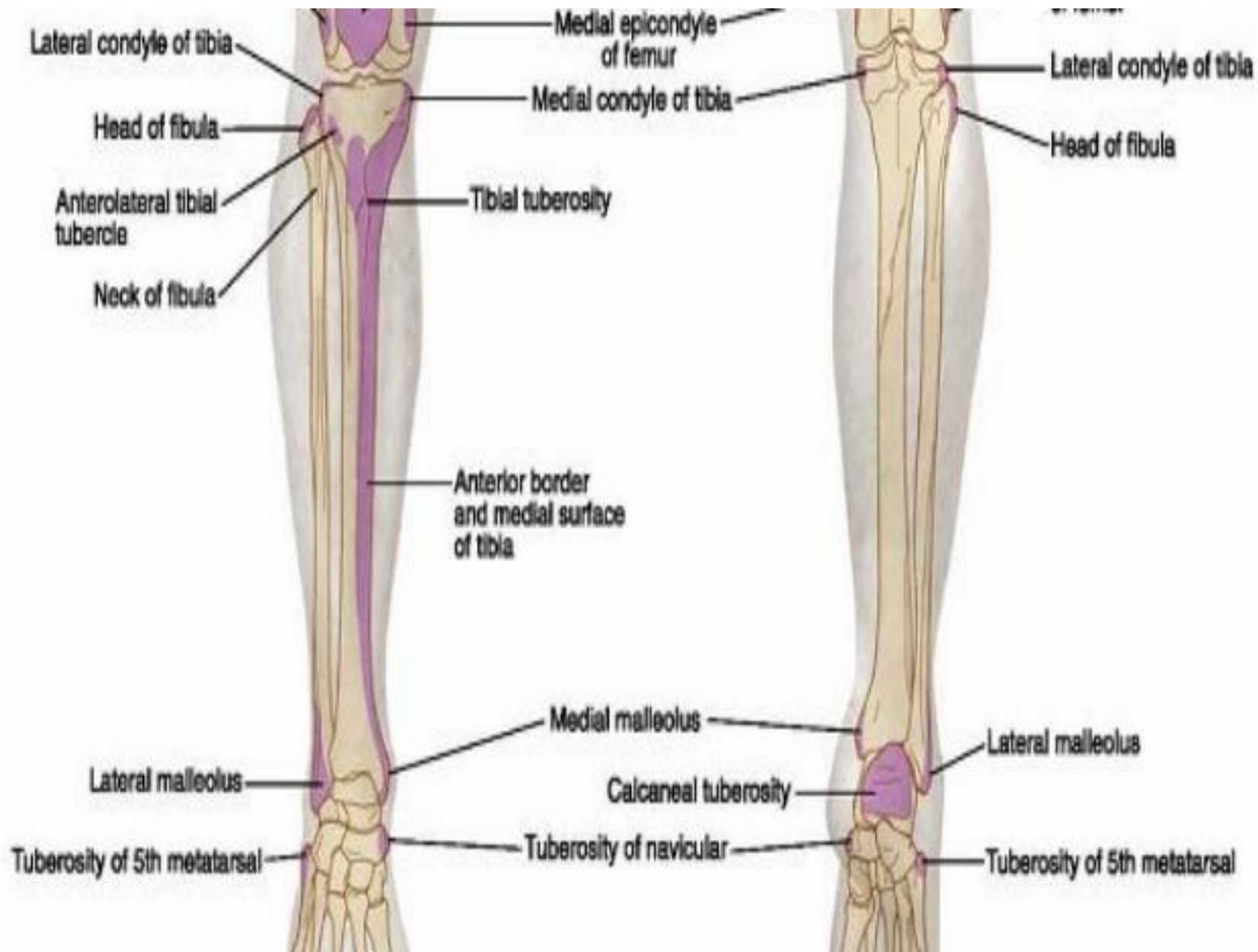
- The following points are palpable
 - Iliac crest
 - Anterior superior iliac spine
 - Posterior superior iliac spine
 - Ischial tuberosity
 - Greater trochanter of femur
 - Femoral condyles
 - Patella
 - Tibial tuberosity
 - Anterior border of tibia
 - Malleoli
 - Head of fibula
 - Tuberosity of navicular
 - Head of first metatarsal
 - Tuberosity of fifth metatarsal

A. Anterior view



B. Posterior view





Superficial structures of lower limb

Fasciae

Fascia lata

- Deep fascia of the thigh
- Strong, investing the limb like an elastic stocking
- Limits outward extension of contracting muscles, making muscular contraction more efficient in compressing veins to push blood toward the heart
- Attaches to and continuous with
 - all around to the root of the lower limb; to the inguinal ligament, pubis, ischium, sacrotuberous ligament, sacrum and coccyx and iliac crest
- Extends downwards over the subcutaneous surface of knee and the tibia (deep fascia of leg)

- It is particularly dense laterally to form iliotibial tract
 - extends from iliac tubercle to tibial tubercle
 - receives the aponeurosis of tensor fasciae latae and gluteus maximus muscles
 - when tensed by its attached muscles, assists in the stabilization of the hip and the extended knee when standing

Gluteal fascia covering gluteus medius (1)

Gluteus maximus (2)

Iliotibial tract

Biceps femoris (3)

Long head

Short head

Gastrocnemius (lateral head) (4)

Tensor of fascia lata (8)

Rectus femoris

Vastus lateralis (7)

Iliotibial tract (6)

Patellar ligament (5)



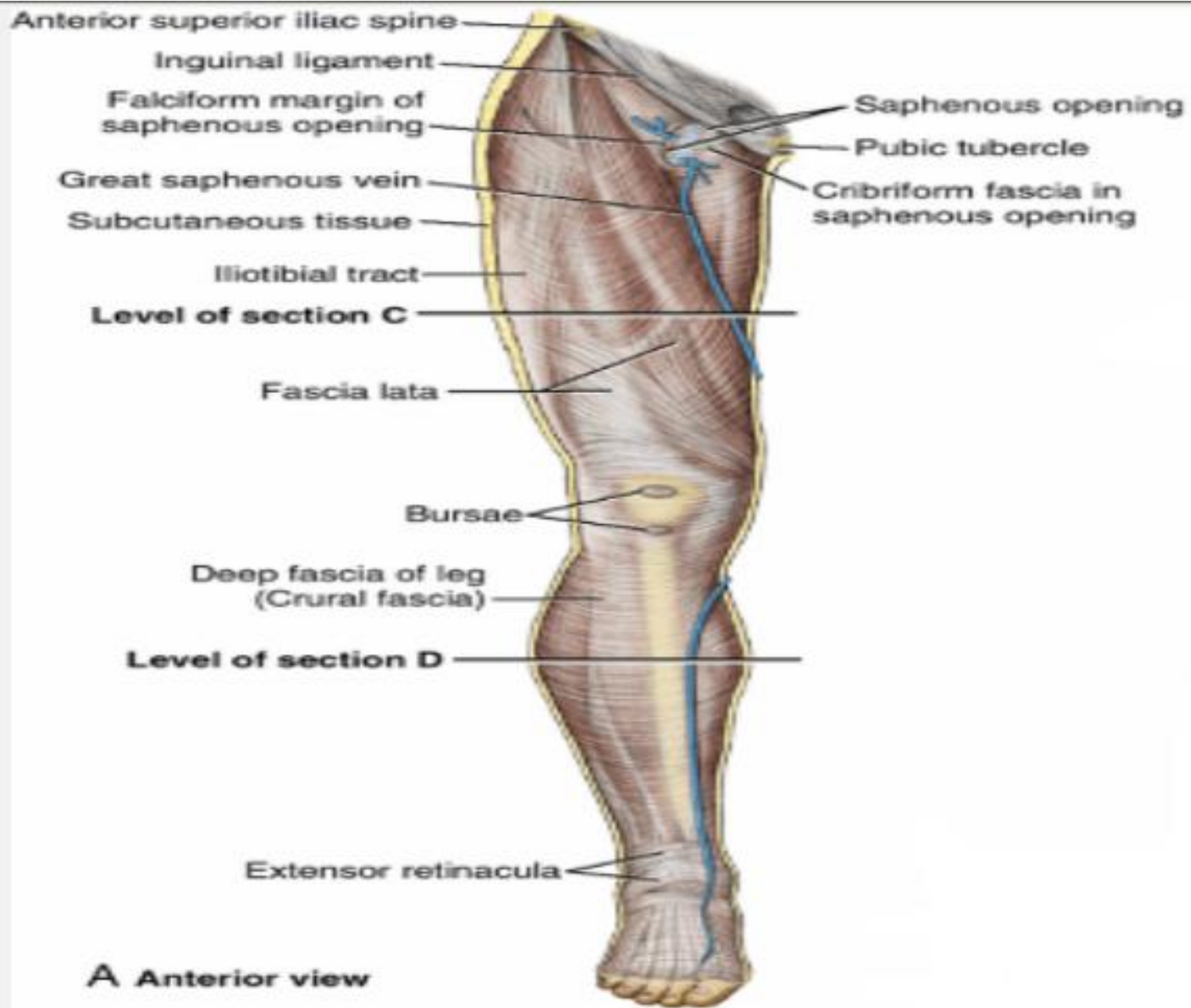
- Fascial compartments
 - 3 intermuscular septa arise from fascia lata and attach to linea aspera
 - Thigh muscles are separated into 3 compartments: anterior, medial and posterior
- Saphenous opening
 - A gap in the fascia lata inferior to medial part of inguinal ligament
 - Great saphenous vein and some lymphatics pass through it

Fascia of the leg (crural fascia)

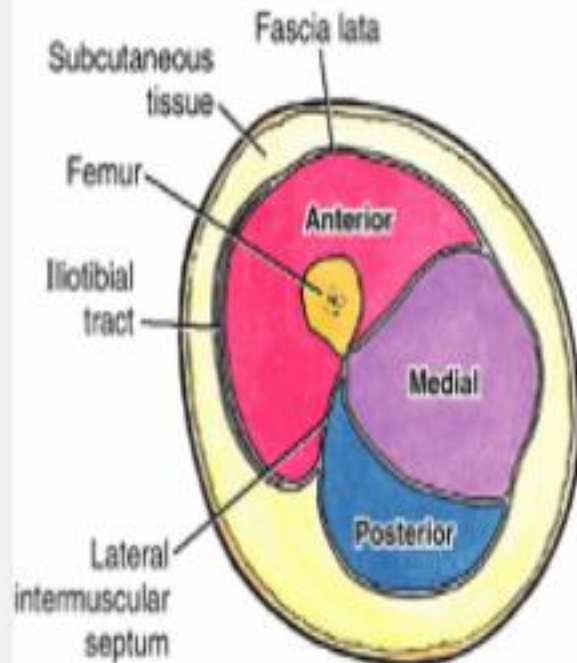
- Continuous with fascia lata
- Attaches to anterior and medial borders of tibia
- Thick proximally and thin distally
- Forms extensor retinacula at distal end
- Anterior and posterior intermuscular septa pass from crural fascia and attach to fibula
- The interosseous membrane and intermuscular septa divide the leg into 3 compartments
 - Anterior (dorsiflexor), lateral (fibular) and posterior (plantarflexor)
 - Transverse intermuscular septum divides the plantarflexors into superficial and deep parts

Plantar fascia

- Attached
 - posteriorly to the medial process of the calcaneus
 - anteriorly to the heads of metatarsals by 5 bands
- Thin in the medial and lateral portions
- Thick centrally as the plantar aponeurosis, which becomes taut when the toes are dorsiflexed



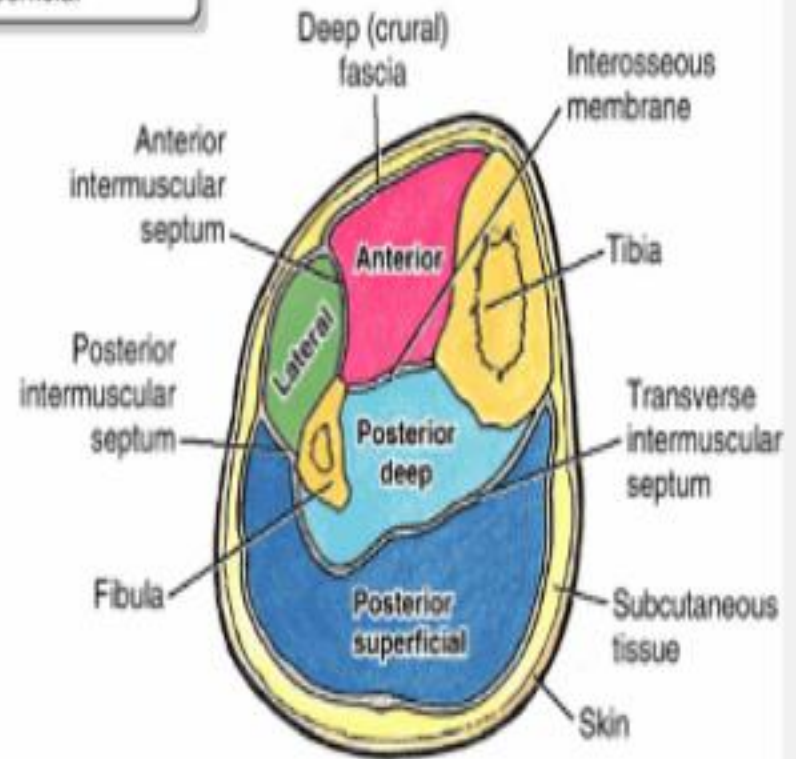
A Anterior view



C Inferior view

Fascial compartments

- Anterior
- Medial
- Lateral
- Posterior deep
- Posterior/posterior superficial



D Inferior view

Venous drainage of lower limb

- Superficial and deep veins
- Superficial veins are in subcutaneous tissue and deep veins accompany arteries

Superficial veins of the lower limb

Great saphenous vein

Origin

- It arises from the union of dorsal digital vein of great toe and dorsal venous arch on the dorsum of the foot

Course

- runs anterior to the medial malleolus
- crosses the lower 1/3 of the medial surface of tibia
- runs 1 cm posterior to the medial border of the tibia

- lies a hand width posterior to the medial border of patella at the knee
- runs up and passes through the saphenous opening and joins with the femoral vein, 4 cm inferolateral to the pubic tubercle
- Anastomoses with the small saphenous vein, below the knee and communicates along intermuscular septa with the deep veins
- Tributaries
 - 3 superficial veins near the inguinal ligament

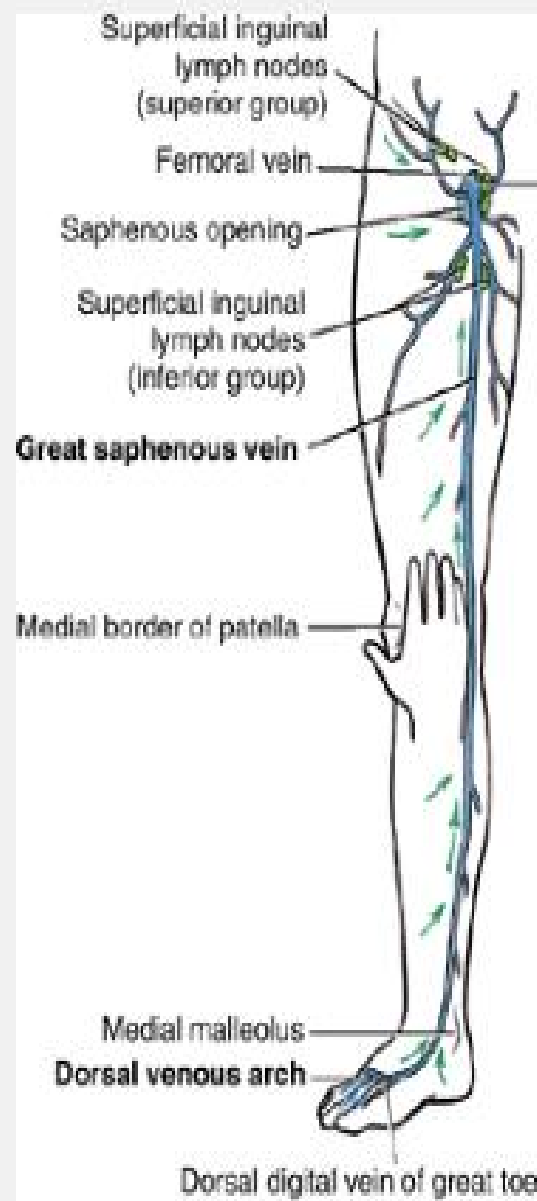
Small saphenous vein

Origin

- Arises on lateral side of the foot from the union of dorsal digital vein of little toe with dorsal venous arch

Course

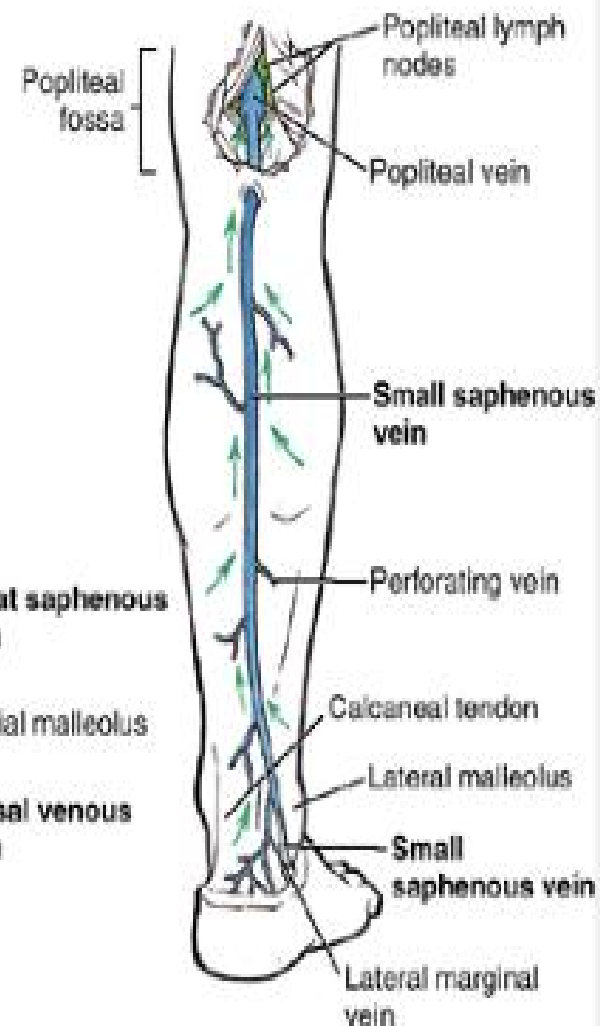
- Ascend posterior to lateral malleolus
- Pass along lateral border of calcaneal tendon
- Incline to midline of fibula and penetrates deep fascia
- Ascends between 2 heads of gastrocnemius muscle
- Empties into popliteal vein in popliteal fossa



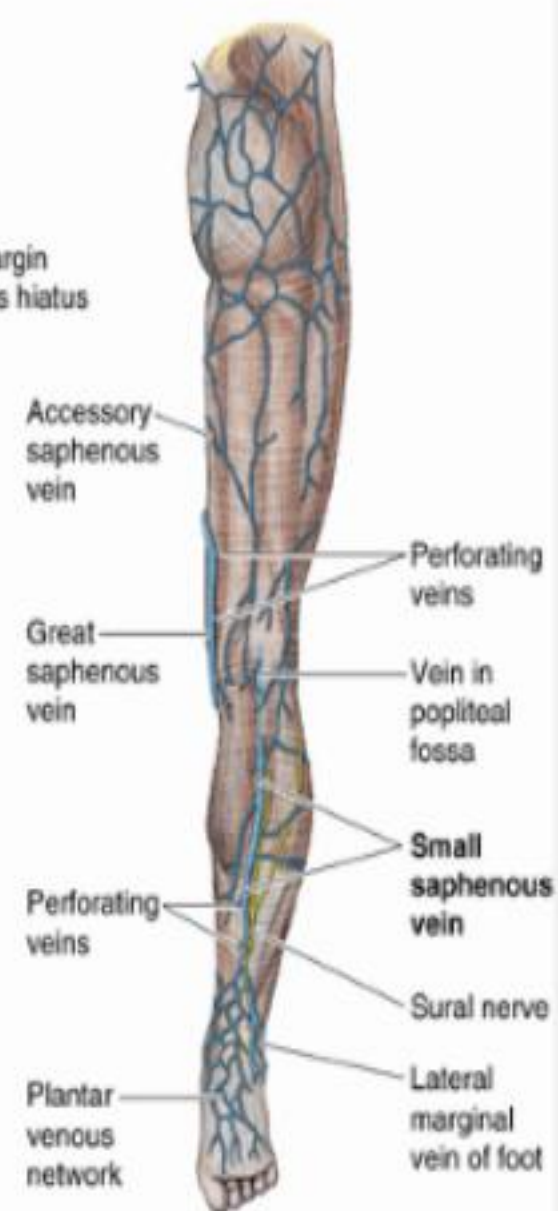
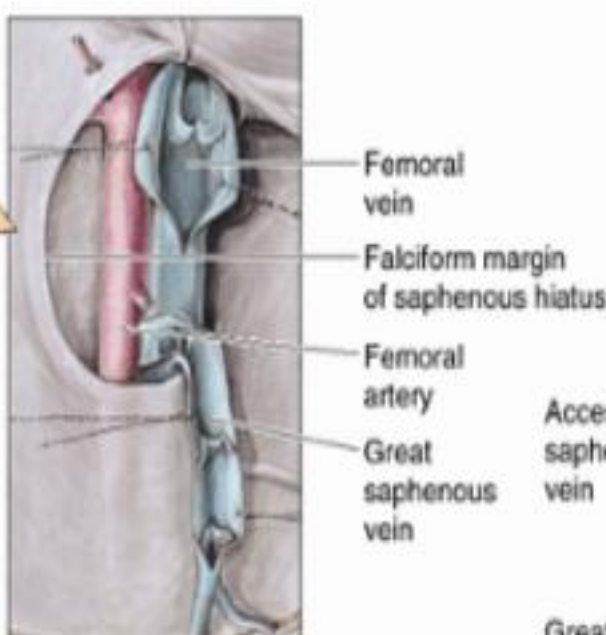
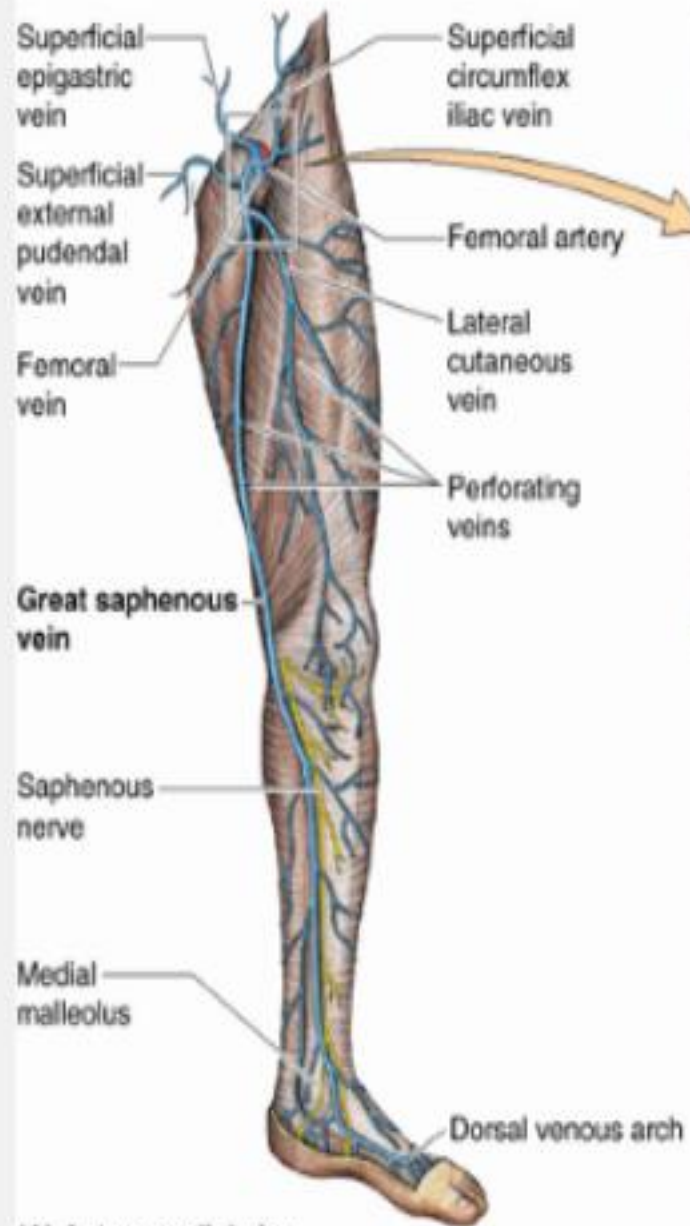
A Anterolateral view



B Anteromedial view



C Posterior view

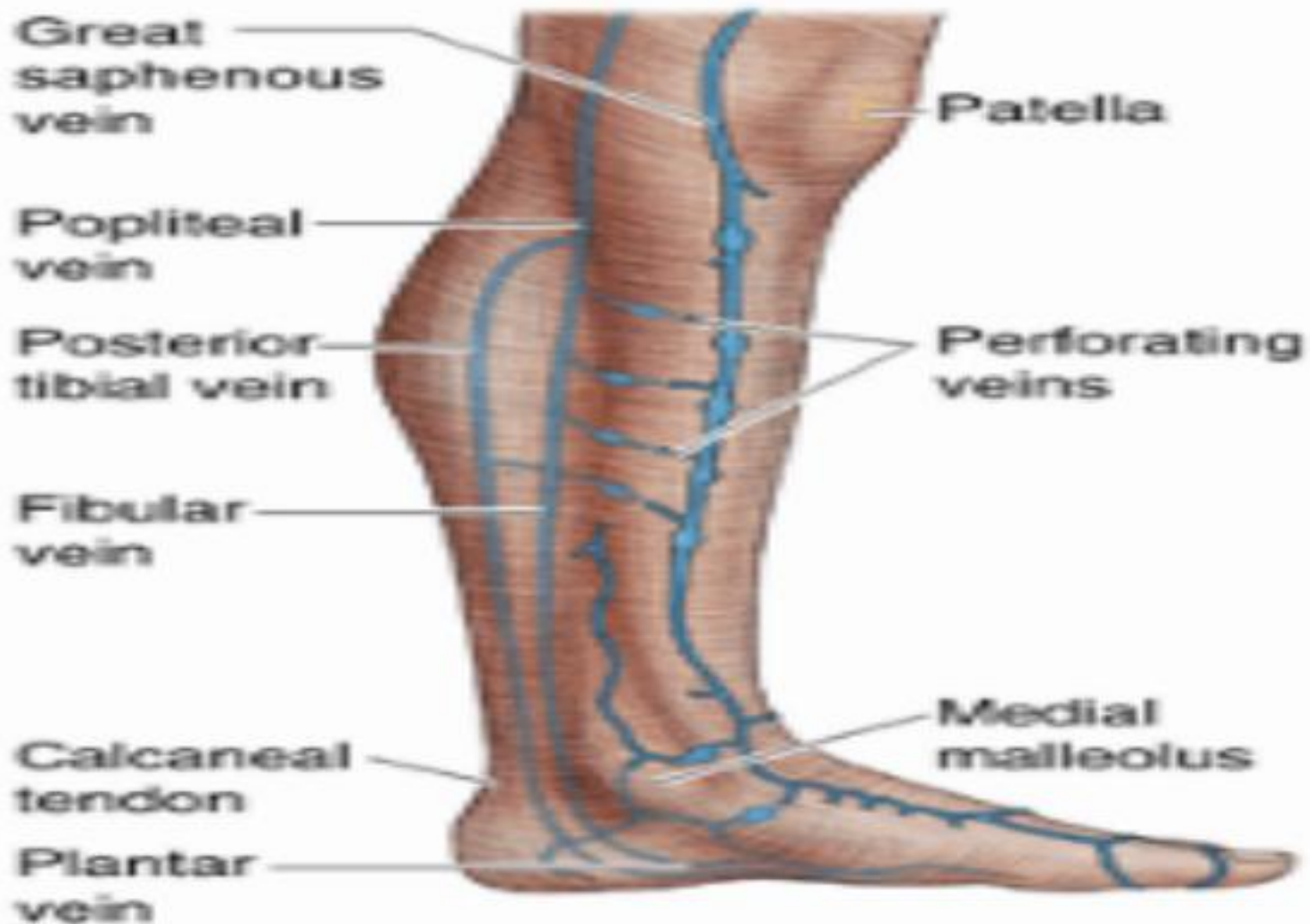


(A) Anteromedial view

(B) Posterior view

Perforating veins

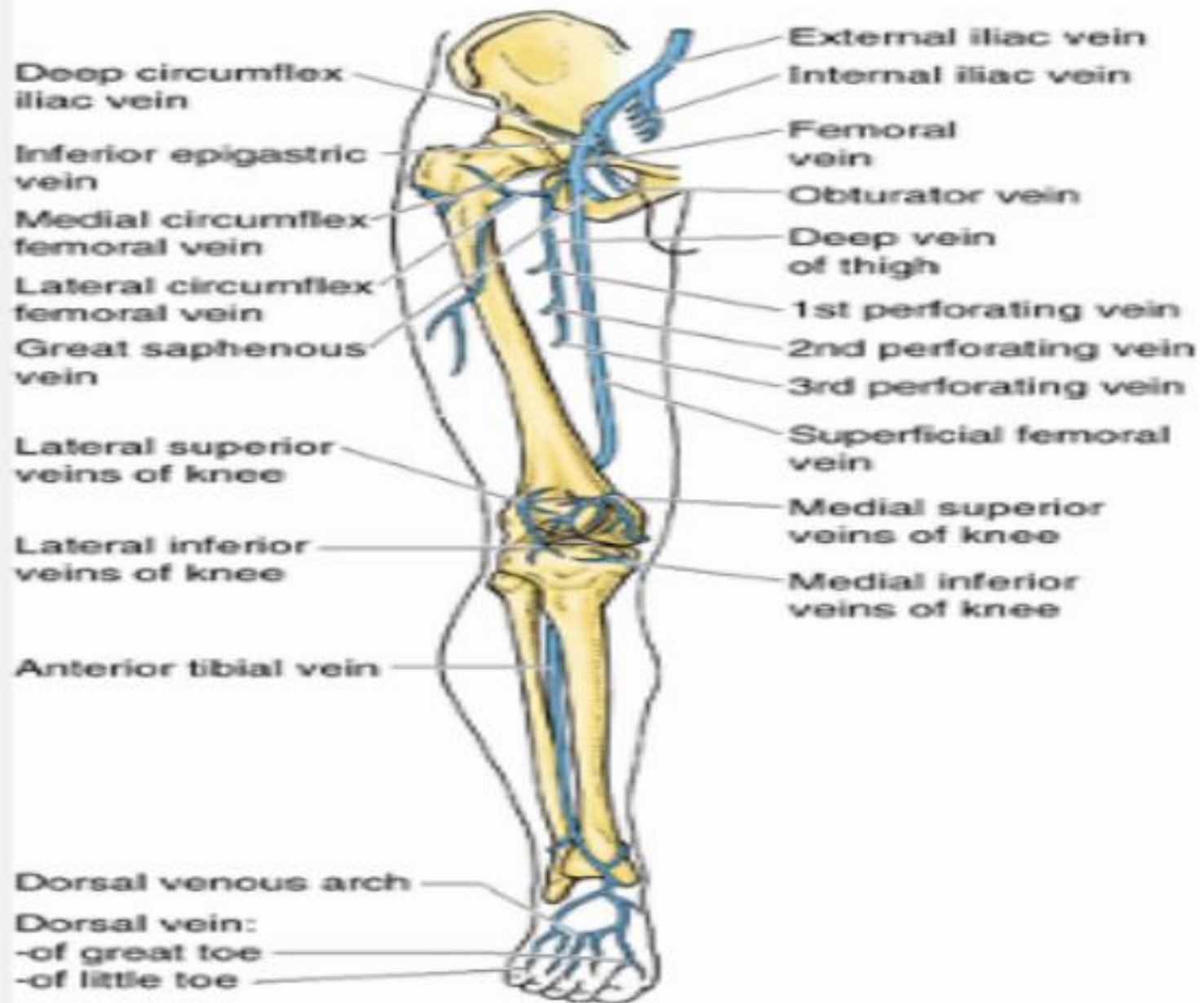
- Originate from superficial veins
- Penetrate deep fascia and join deep veins
- Contain valves that allow blood to flow from superficial veins to deep vein
- This pattern is important for venous return because it enables muscular contractions to propel blood toward the heart



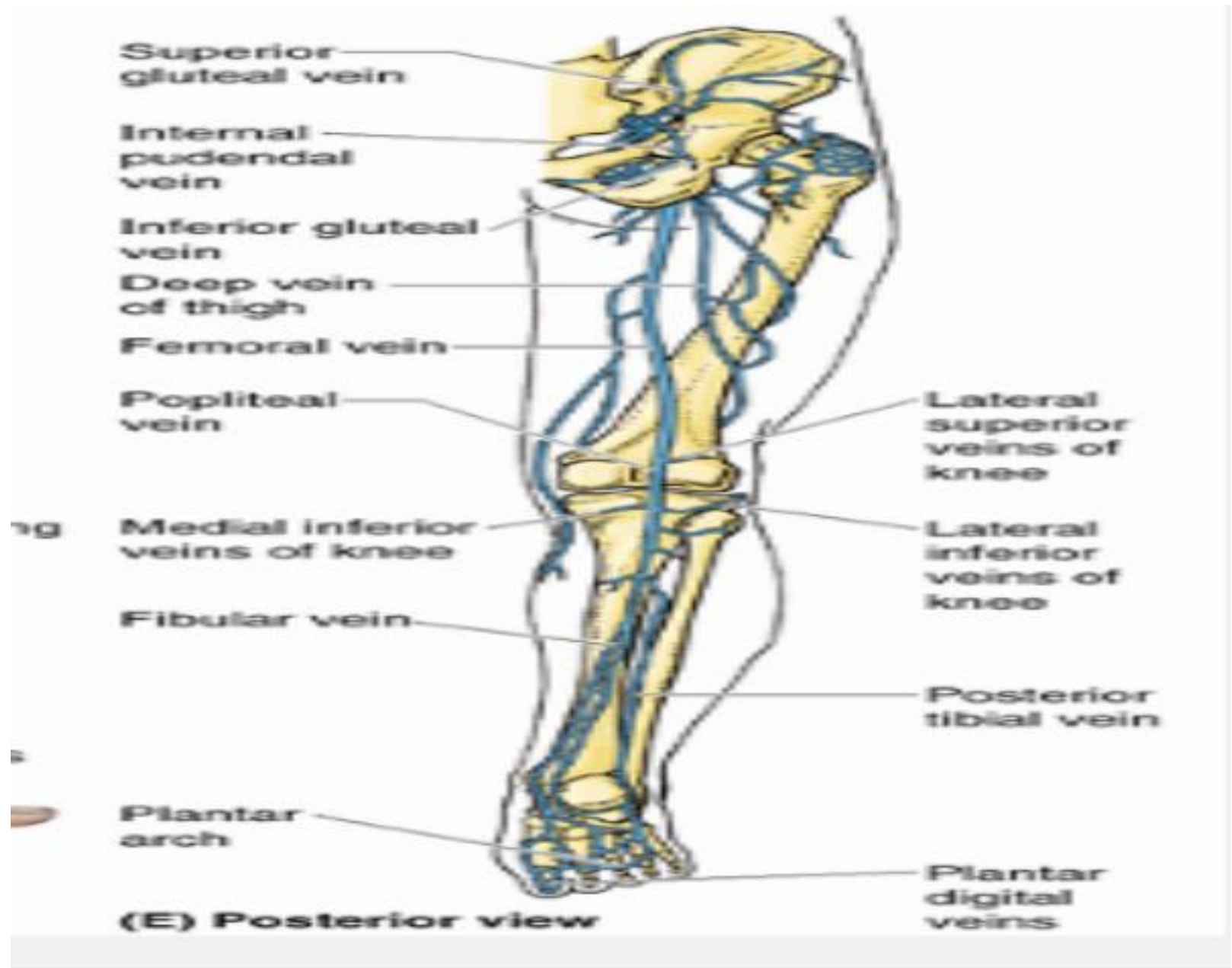
(D) Medial view

Deep veins of the lower limb

- Accompany arteries
- Usually paired, frequently interconnecting veins that flank the artery
- Contained within vascular sheath with the artery
 - arterial pulsations help move the blood
- In the leg they flow to popliteal vein, which become femoral vein in thigh which become external iliac vein in pelvis



(C) Anterior view



Lymphatic drainage of lower limb

- Superficial and deep lymph vessels
- Superficial lymph vessels
 - There are many than the deep vessels
 - Many of them follow the great saphenous vein and end in the superficial inguinal lymph nodes
 - Few of them follow the small saphenous vein and end in to popliteal lymph nodes
- Deep lymph vessels of the leg
 - follow the deep veins and enter popliteal lymph nodes from which lymph drain into the deep inguinal lymph nodes

Lymph nodes

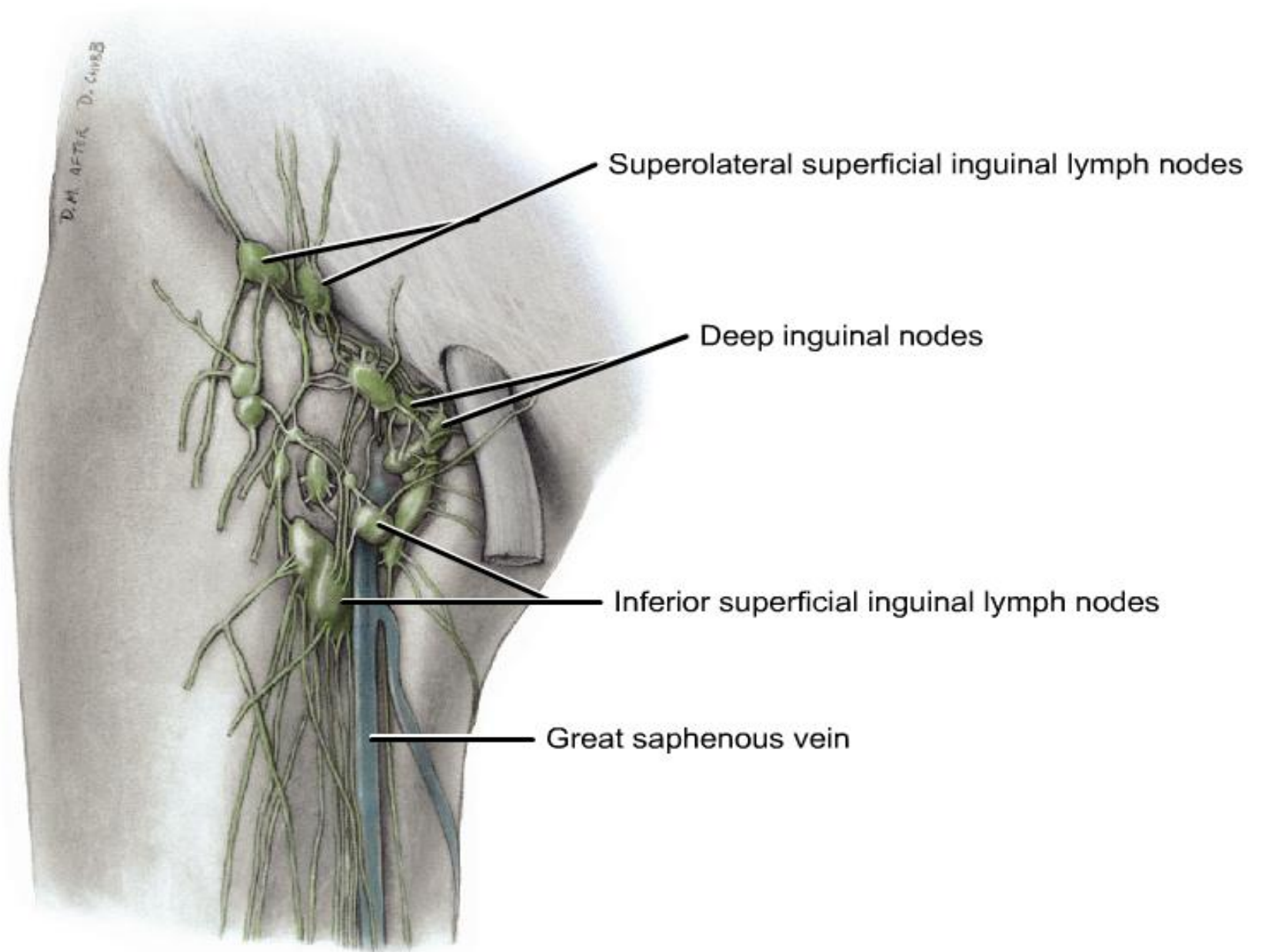
Inguinal lymph nodes

- palpable
- divided into superficial and deep groups

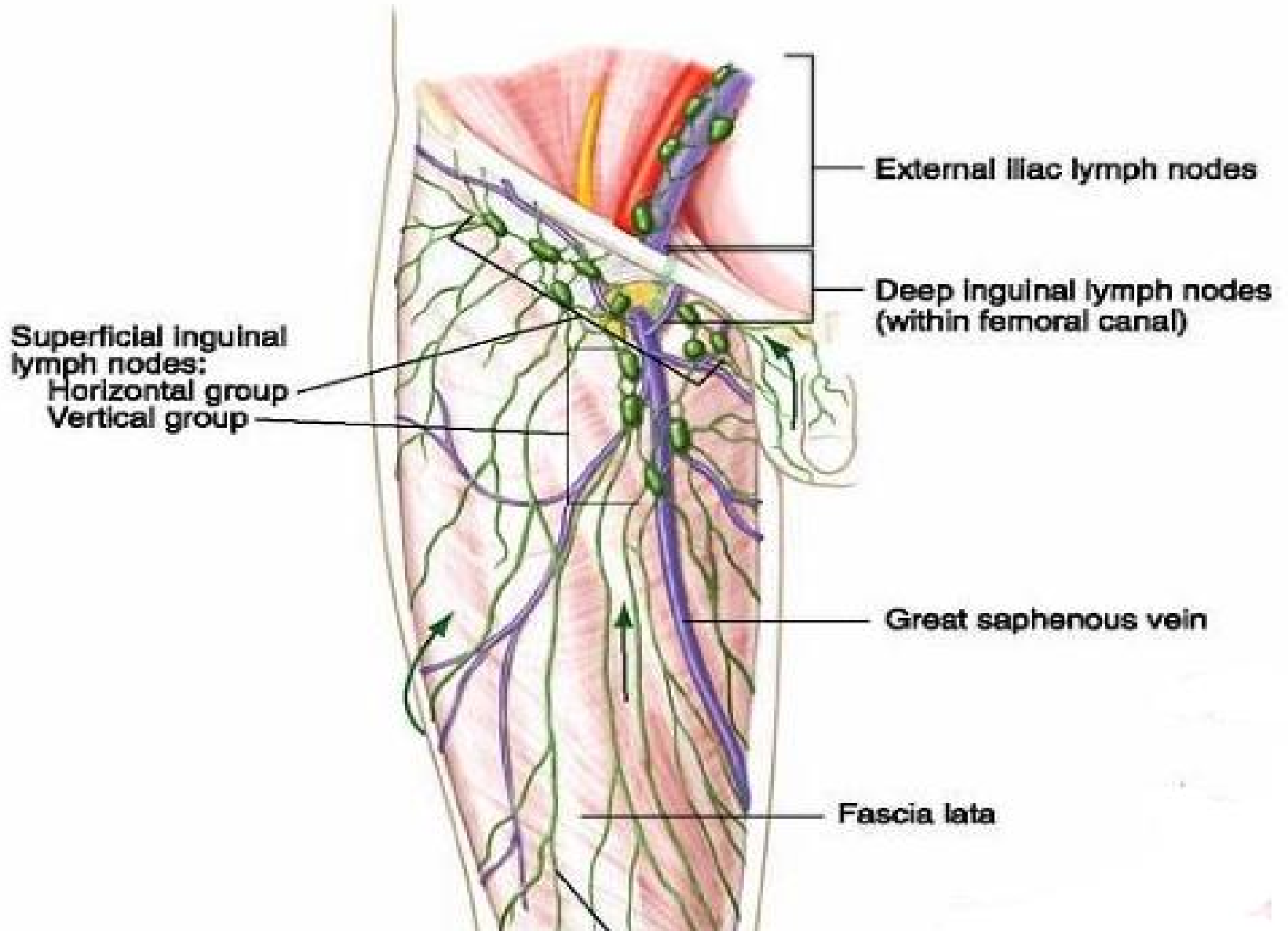
Superficial inguinal lymph nodes

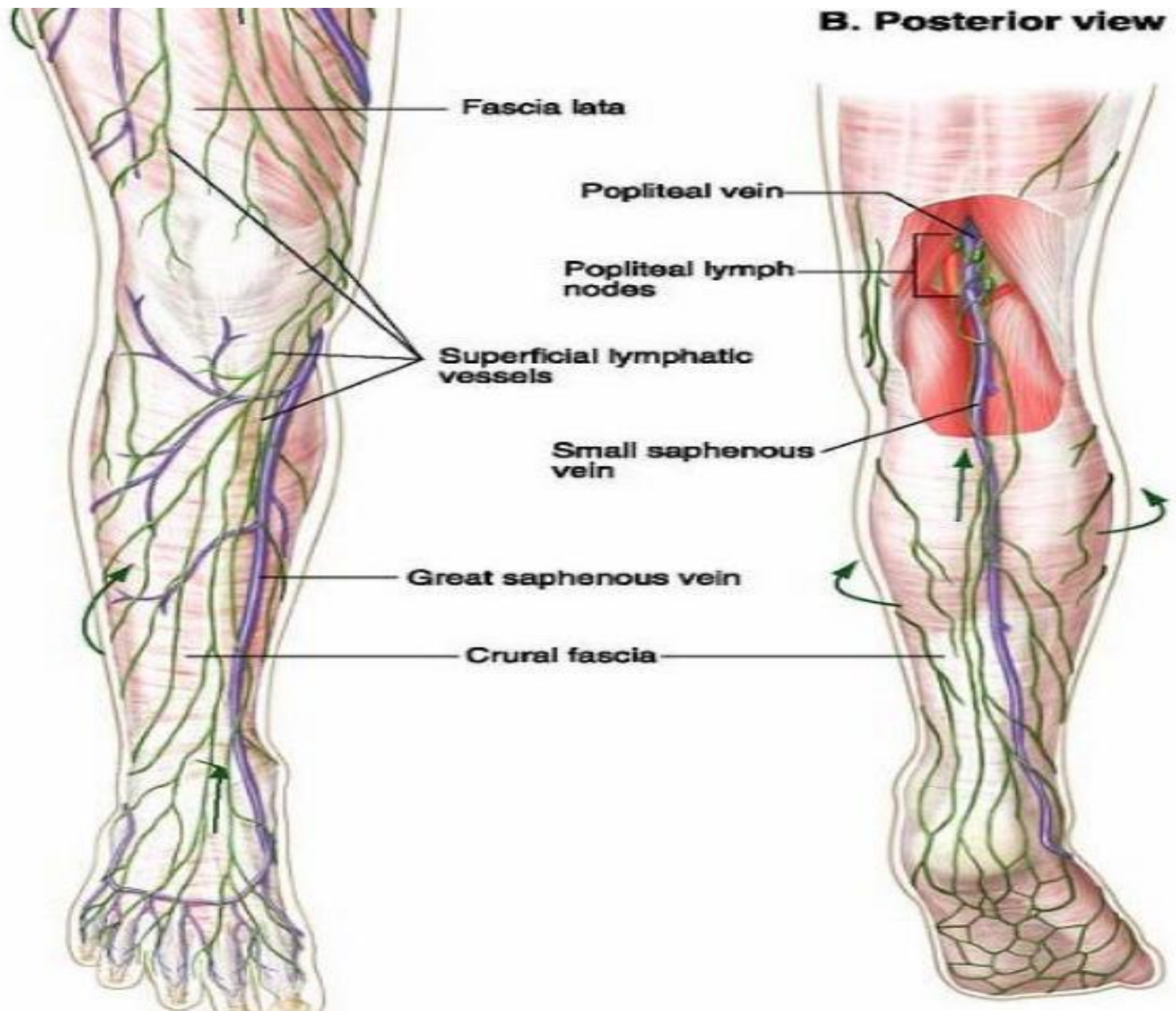
- divided into
 - **superior horizontal** (superolateral and superomedial) group
 - drain
 - subcutaneous tissues of the anterior abdominal wall inferior to the umbilicus
 - penis and scrotum, or the vulva
 - distal part of the vagina
 - gluteal region, perineum
 - distal part of the anal canal
 - **inferior vertical** group
 - lies along the great saphenous vein
 - receives most lymph vessels of the lower limb

- Deep inguinal lymph nodes
 - Receive efferent vessels from superficial nodes
 - Efferent from the deep inguinal lymph nodes ascend in the femoral canal and deep to the inguinal ligament to end into the external iliac lymph nodes



A. Anterior view





Clinical correlates

- Varicose veins
 - form when the valves that usually prevent blood flow from the deep veins through the perforating veins to the superficial veins are incompetent
 - As a result, the superficial veins become varicose (dilated so that their cusps do not close) and tortuous



Medial view

- Thrombosis and thromboembolism
 - The veins of the lower limb are subject to venous thrombosis (blood clotting) for example, after a bone fracture, muscular inactivity, and external pressure on the veins (tight cast)
 - Venous stasis (stagnation) is an important cause of thrombus formation
 - A thrombus that breaks free from a lower limb vein and travels to the lungs results in a pulmonary thromboembolism (obstruction of the pulmonary artery)

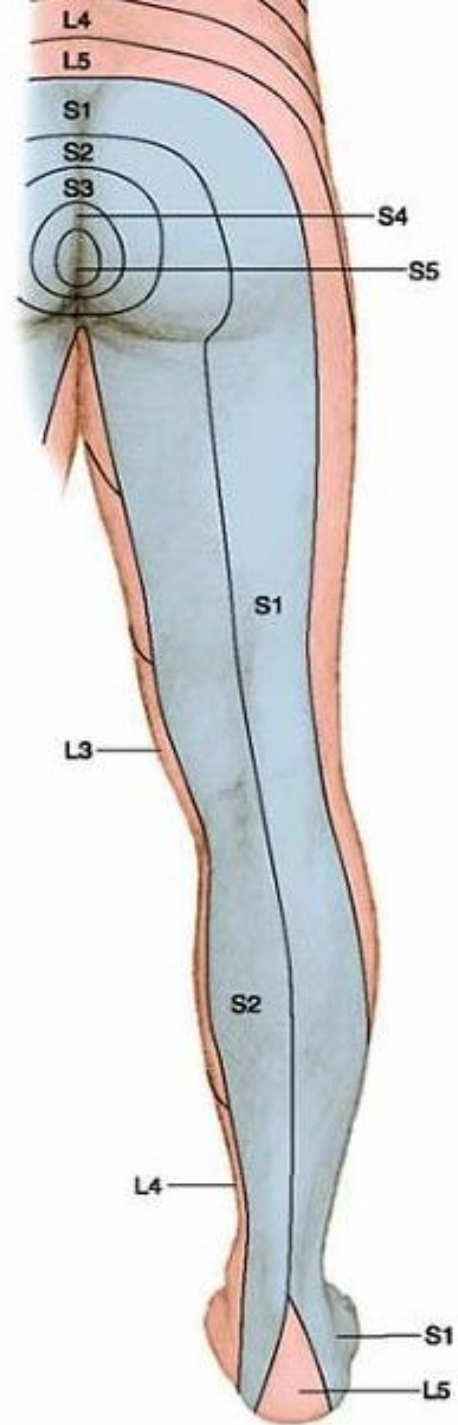
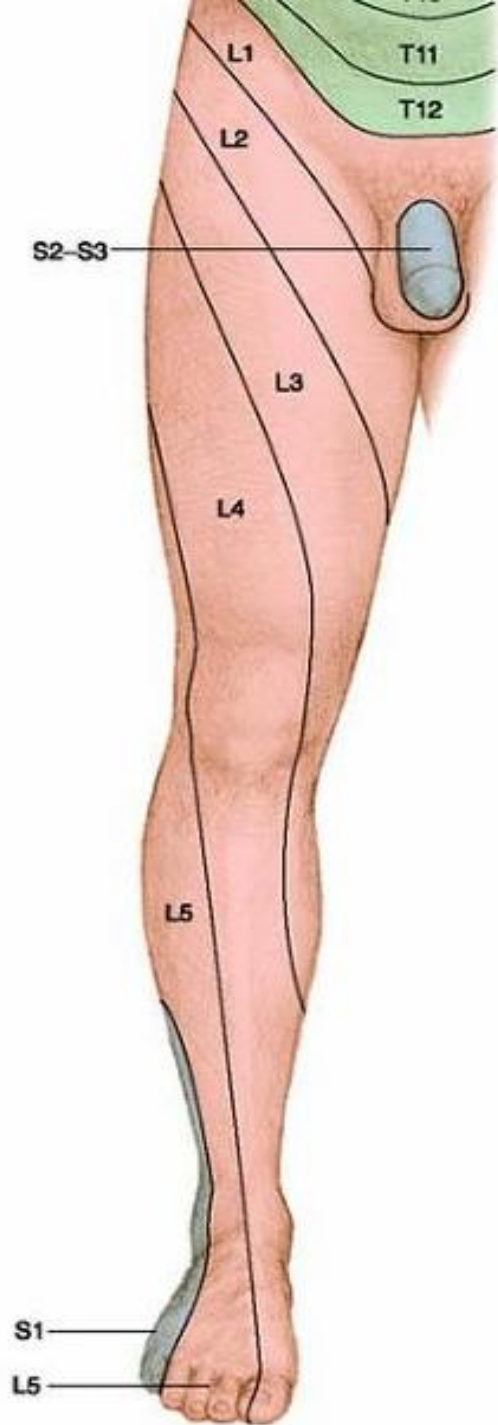
- Saphenous vein grafts
 - Vein grafts obtained by surgically harvesting parts of the great saphenous vein are used to bypass obstructions in blood vessels (e.g., an occlusion of a coronary artery or its branches)
 - When part of the vein is used as a bypass, it is reversed so that the valves do not obstruct blood flow
 - Because there are so many other leg veins, removal of the great saphenous vein rarely affects circulation seriously, provided the deep veins are intact

- Enlarged inguinal lymph nodes
 - Causes
 - pathogenic microorganisms, malignancies and perineal abscess

Cutaneous innervation of lower limb

- Cutaneous nerves arise from lumbar and sacral plexuses
- They pierce the deep fascia along the upper border of the Sartorius
- Those arising from the femoral nerve form the anterior femoral cutaneous nerve (the intermediate and medial cutaneous nerves of the thigh)
- Others arise from the saphenous nerve

- Arise from 10 spinal cord segments – T12- S4
 - Upper and lateral gluteal region - T12 & L1,2
 - Front, medial and lateral thigh - L1,2,3
 - Medial and front of the leg - L 4
 - Lateral and front of the leg and intermediate of the dorsum of the foot - L5
 - Lateral border of the foot - S1
 - Sole of the foot medial part - L4
 - Sole of the foot intermediate part - L5
 - Sole of the foot lateral part - S1
 - Back of the leg and thigh - S2
 - Inferior and medial part of gluteal region and perianal region - S3,4



Cutaneous nerves

- Hip region
 - iliohypogastric and middle clunial for lateral and medial regions respectively
- Thigh
 - Lateral - lateral cutaneous nerve of the thigh
 - Superolateral - iliohypogastric
 - Superoventral - genitofemoral
 - Superomedial - ilioinguinal
 - middle of the medial part - obturator
 - Ventromedial - medial femoral cutaneous
 - Ventrolateral - intermediate femoral cutaneous
 - Posterior - posterior cutaneous nerve of the thigh

- Leg
 - Ventromedial – saphenous
 - Lateral and superior - lateral cutaneous nerves of the calf
 - Ventral and inferior - superficial peroneal
 - Lateral and inferior - sural

- Foot
 - Dorsum
 - Lateral - sural
 - Medial - saphenous
 - Intermediate - superficial peroneal
 - Adjacent areas of the 1st and 2nd digits - anterior tibial
 - Sole
 - Medial side and skin of the medial 3 & ½ toes - medial plantar
 - Lateral side of skin over the lateral 1 & ½ toes - lateral plantar

